Test-Time Context Encoding: Store and Use Context in LLM Weights

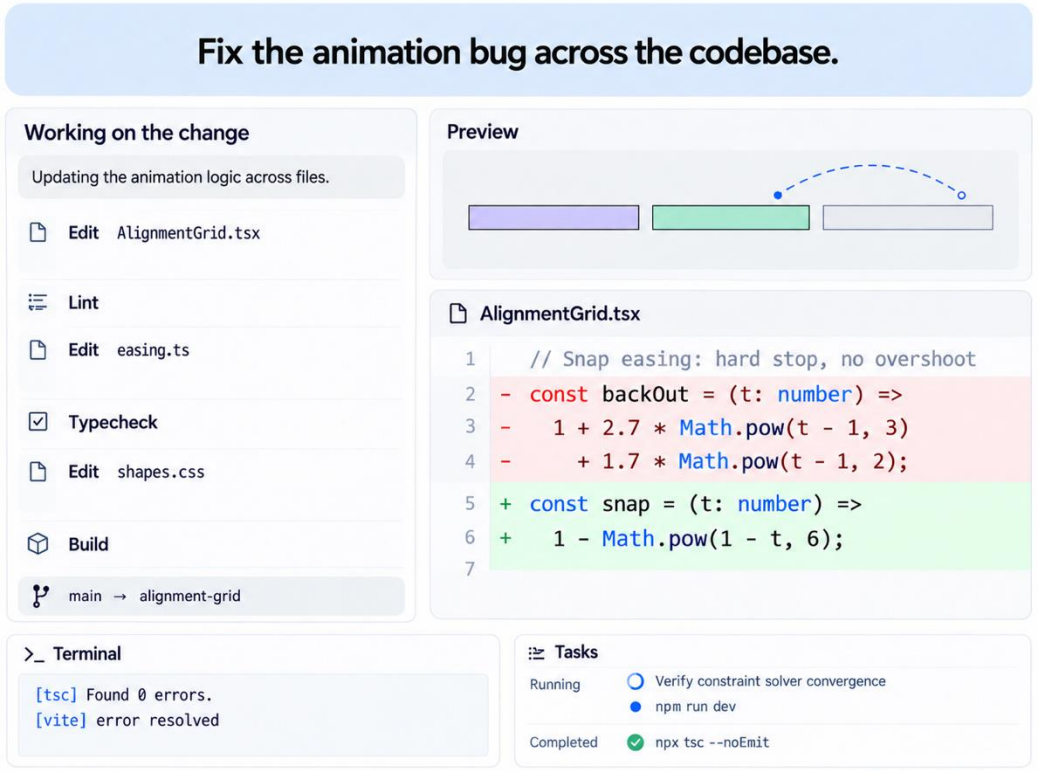
Qizhou WANG
Imperfect Information Learning Team

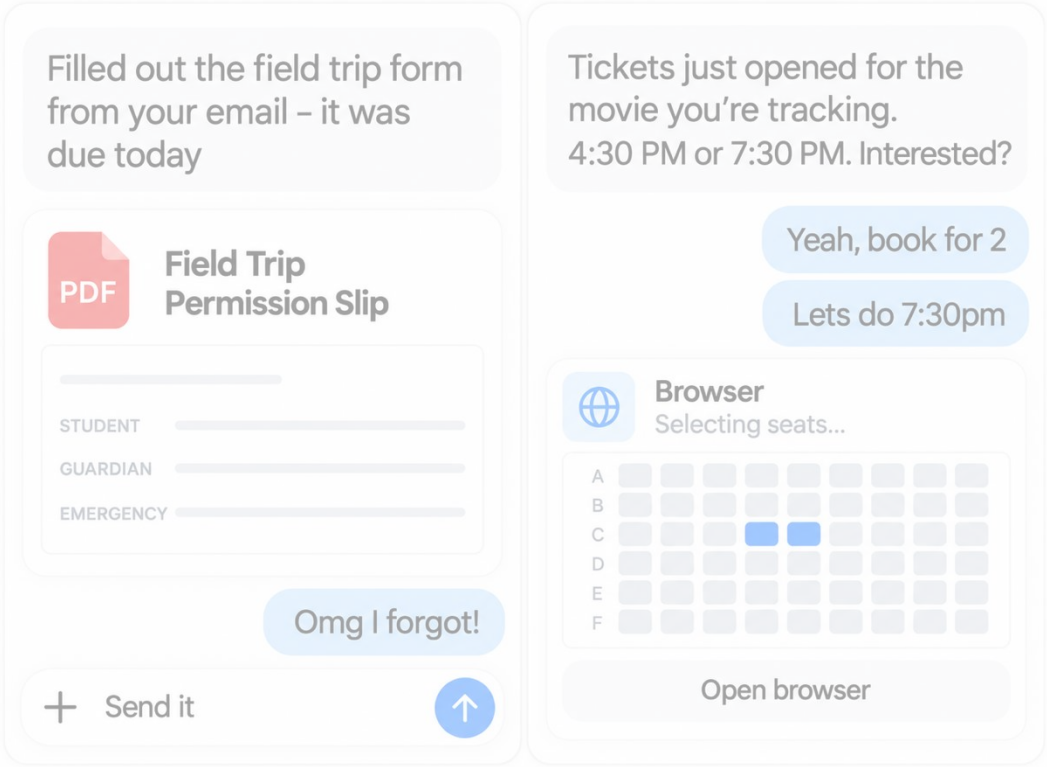
https://qizhouwang.github.io/homepage





LLMs are taking on more complex tasks

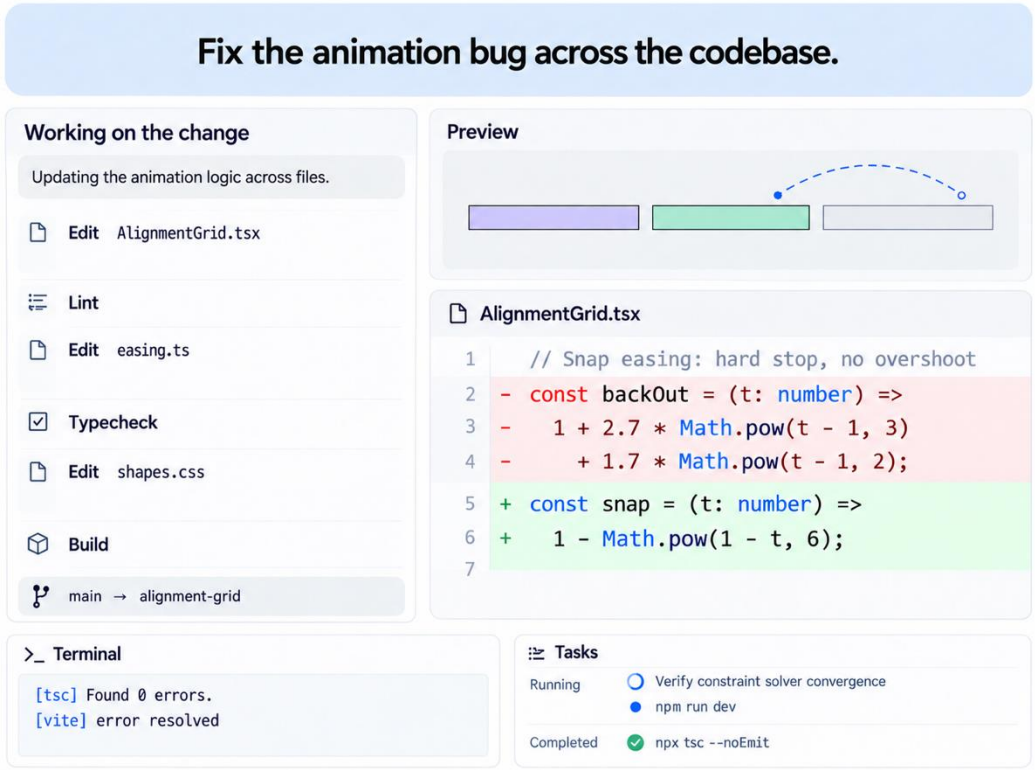


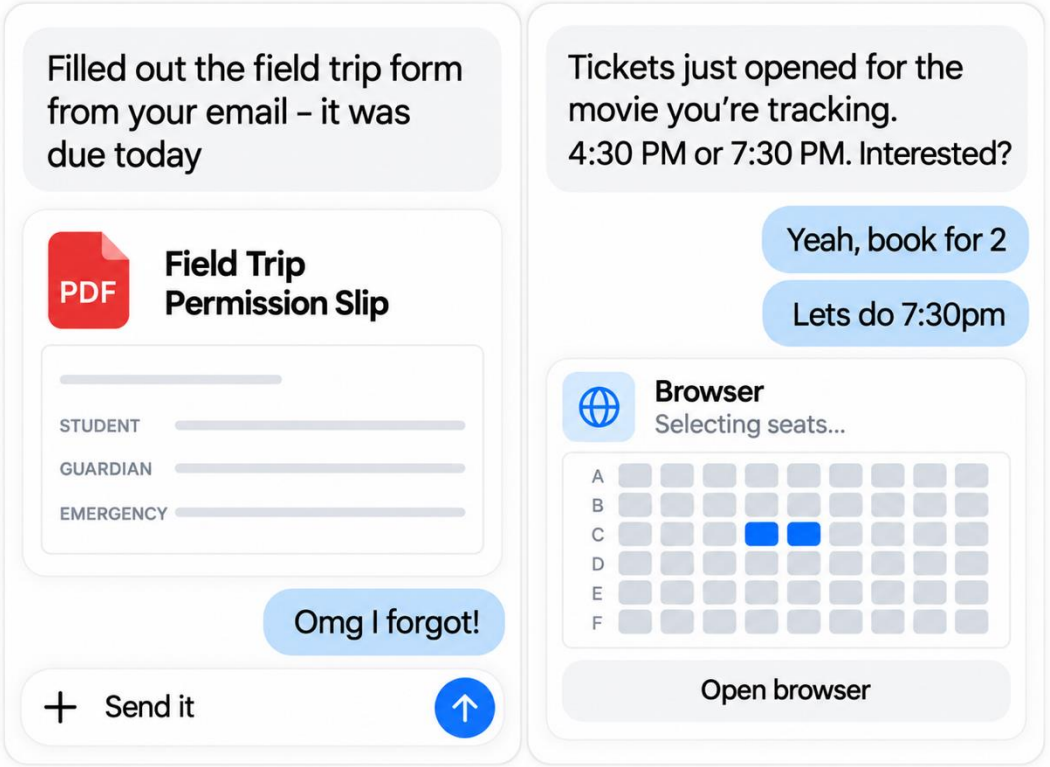


Coding. Coordinating code changes across large and complex repositories.

Personal Assistant. Managing everyday tasks across weeks of interactions.

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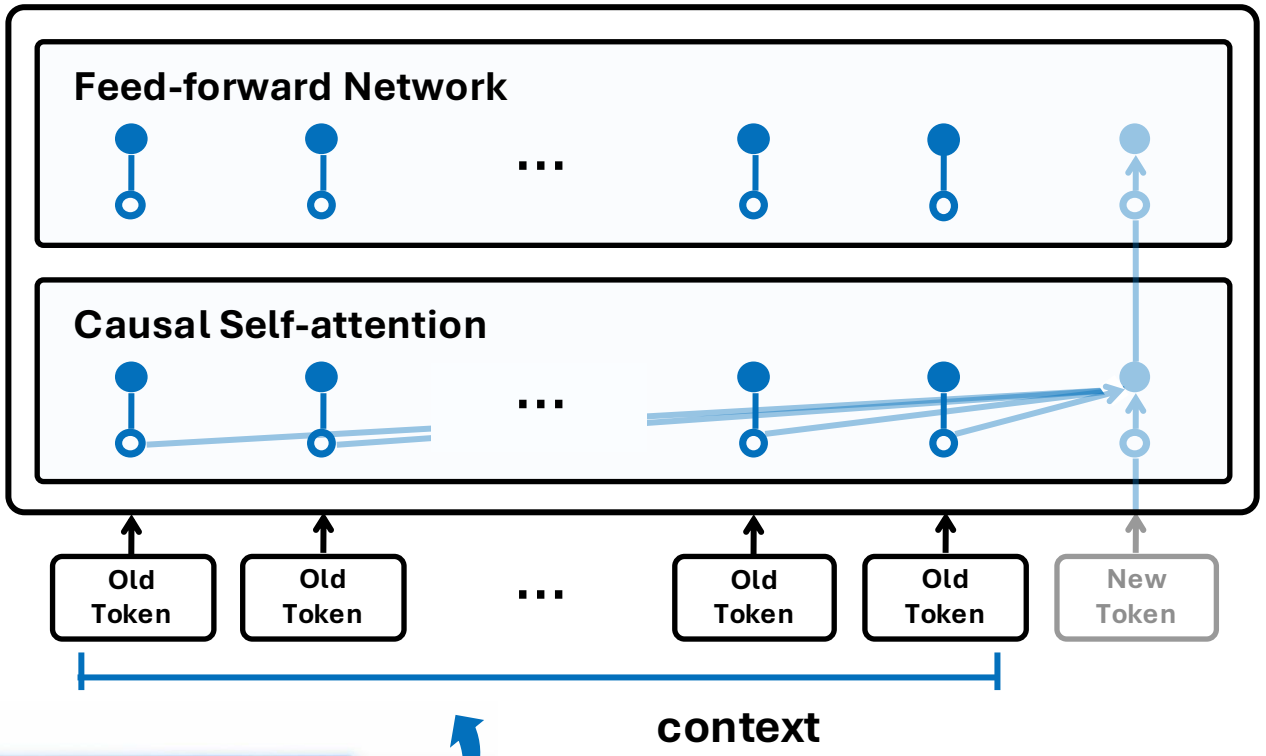


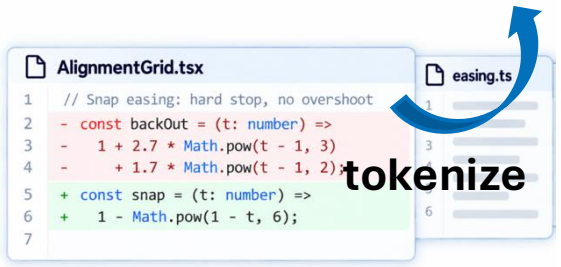


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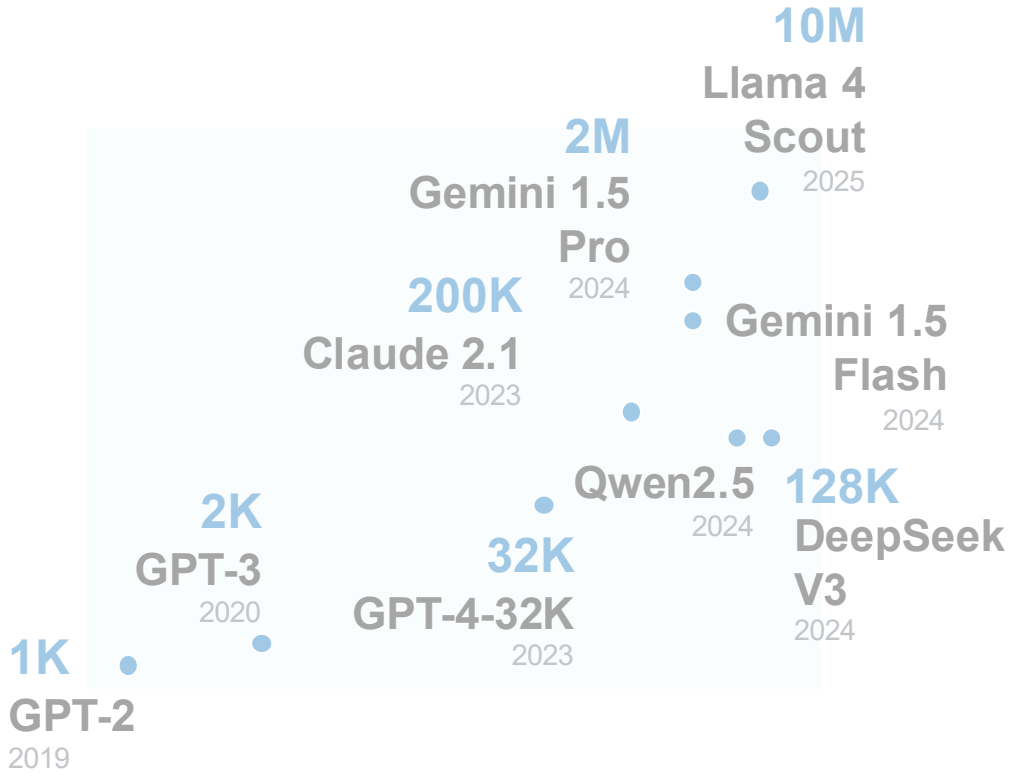
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LLMs can handle increasingly more tokens



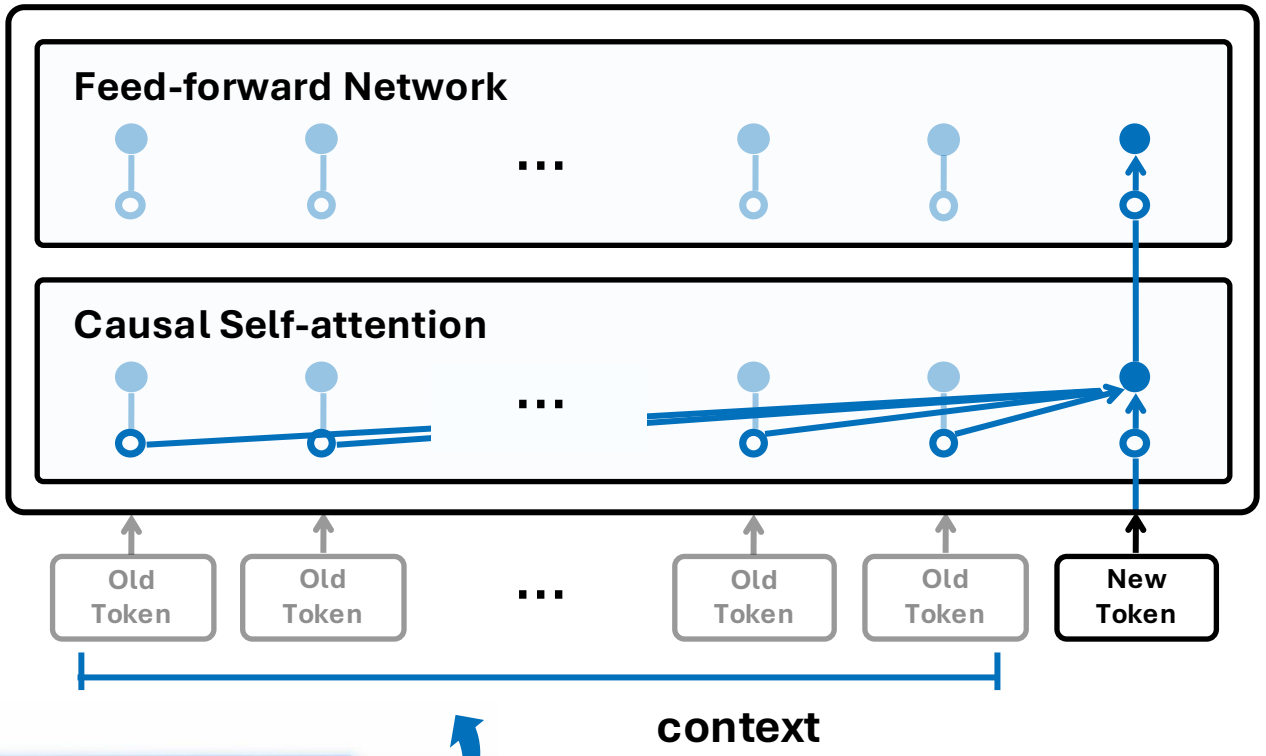


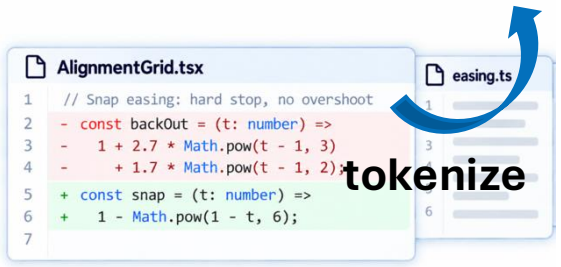
More tokens for harder tasks and longer interactions.



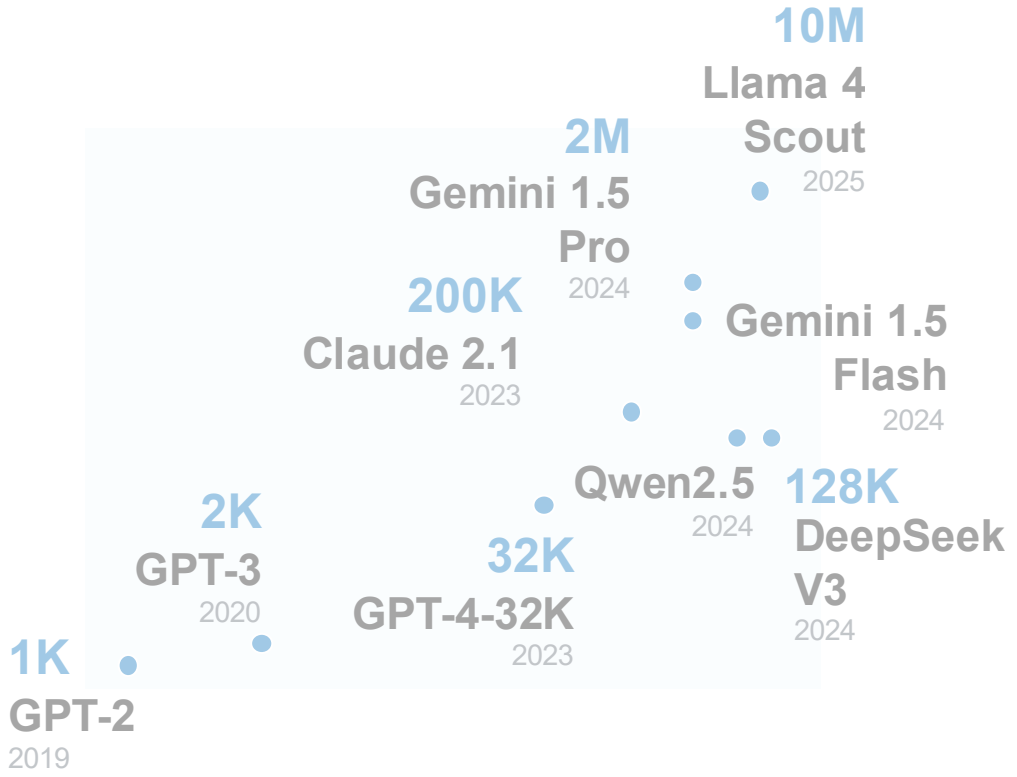
Growth in the token capacity of advanced LLMs over years.

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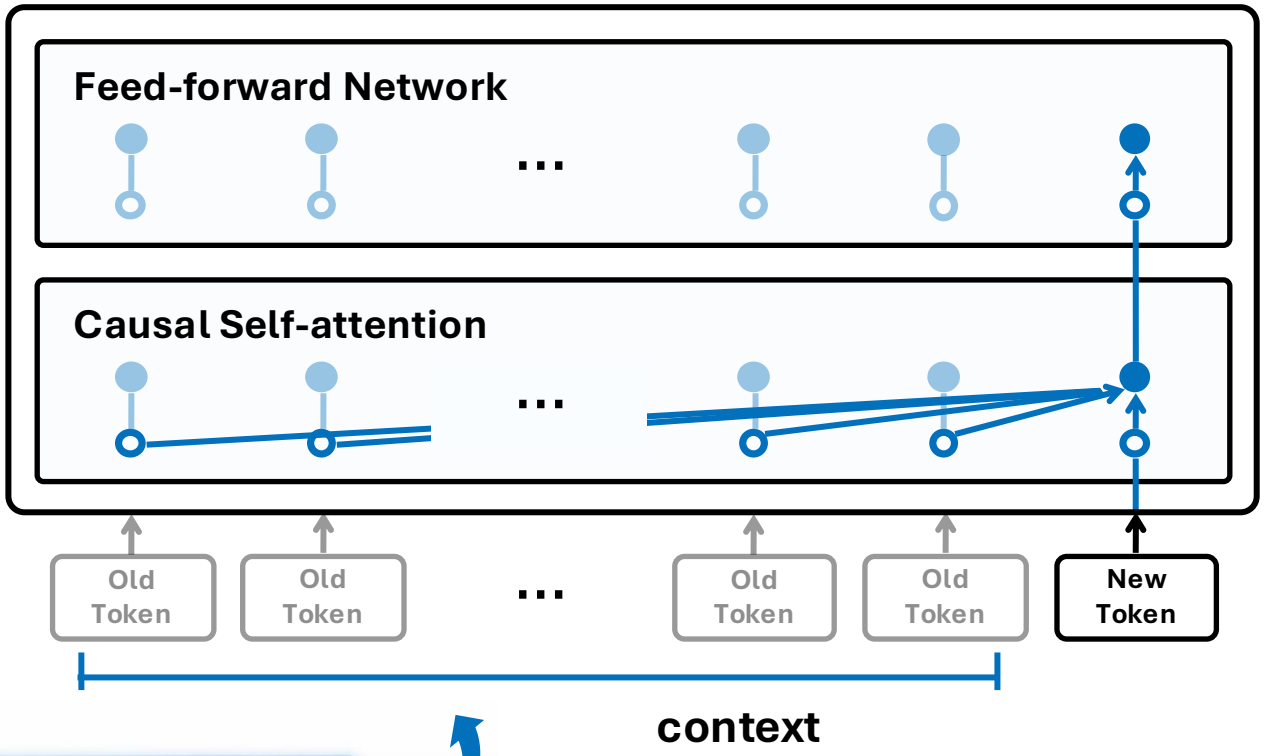


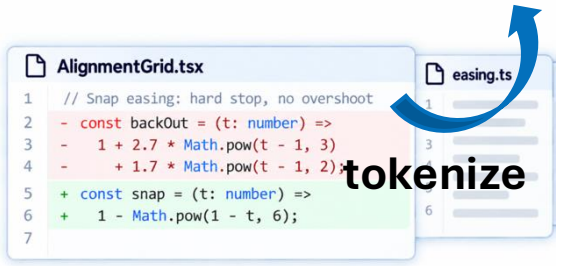
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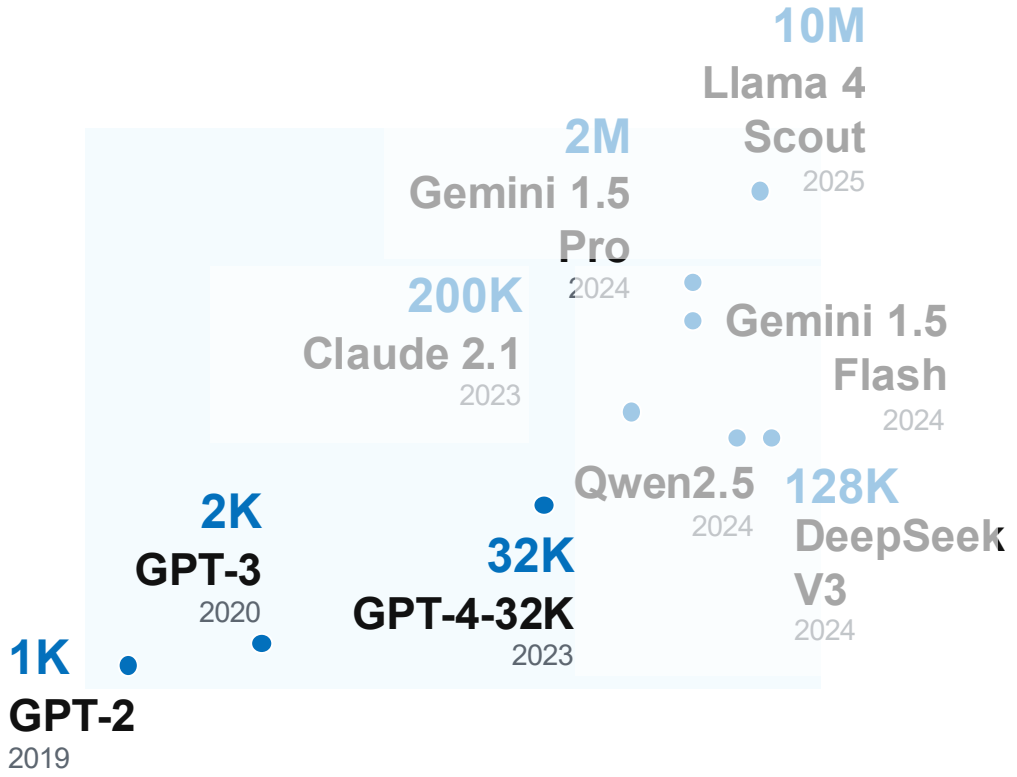
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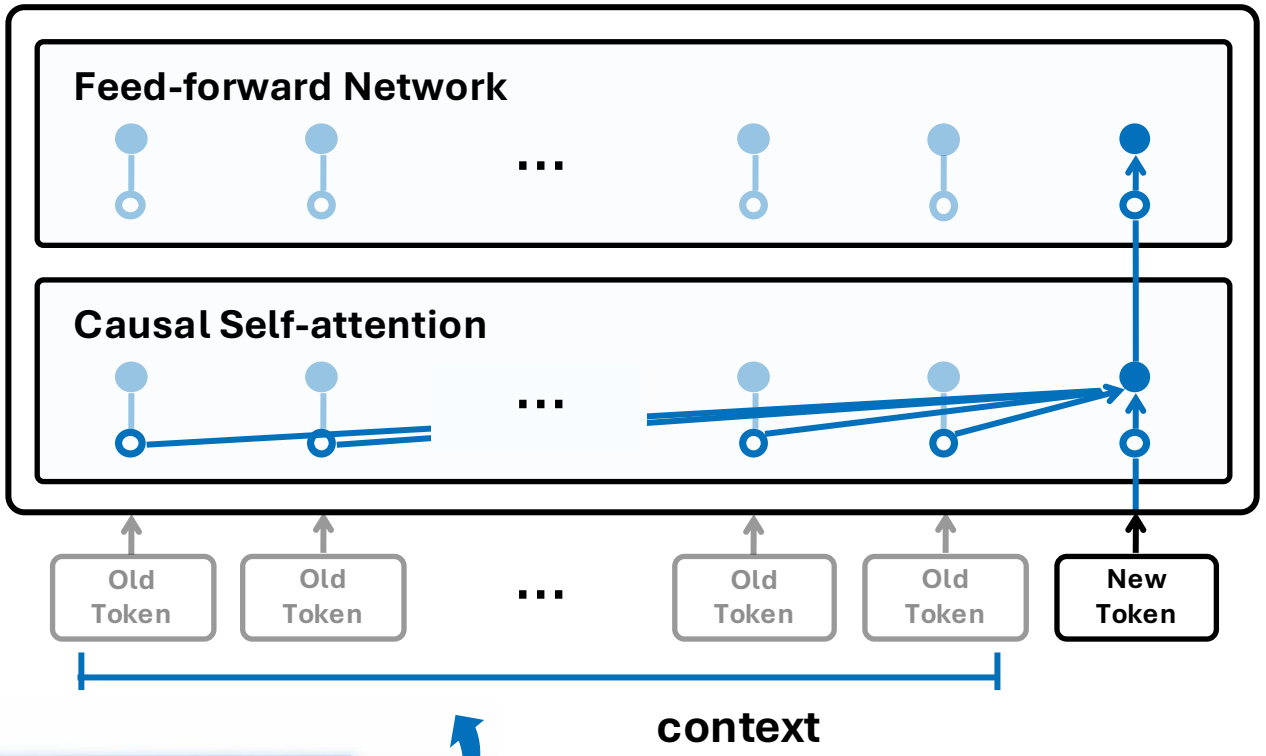


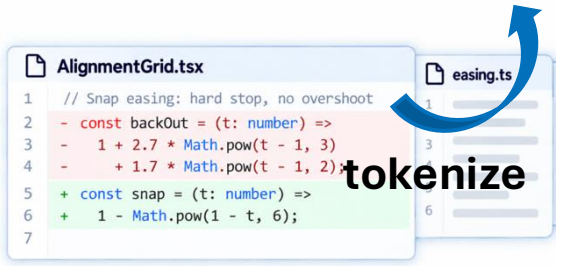
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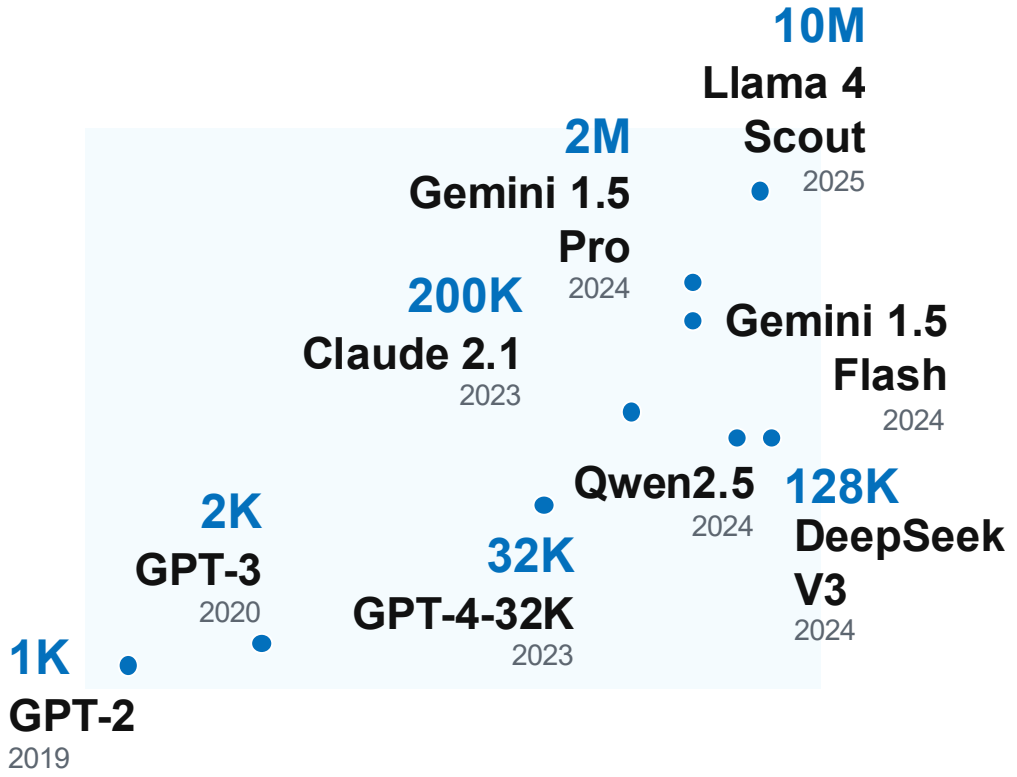
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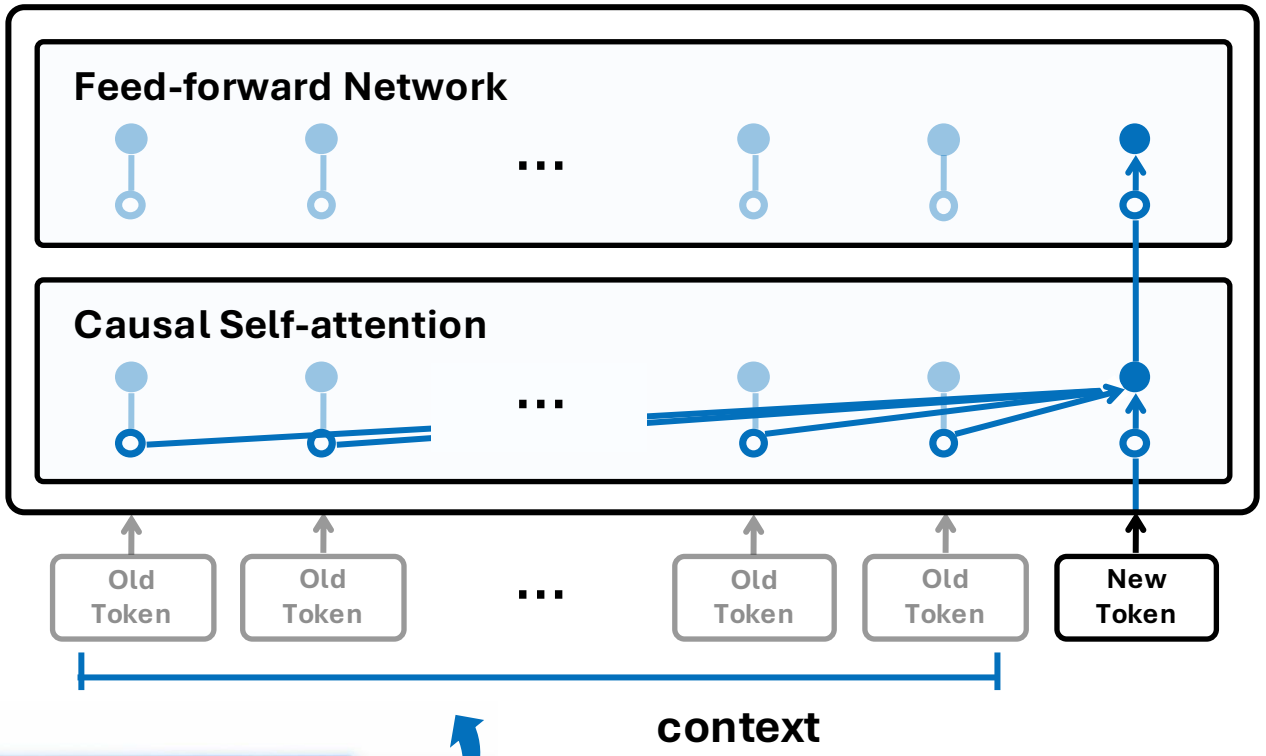


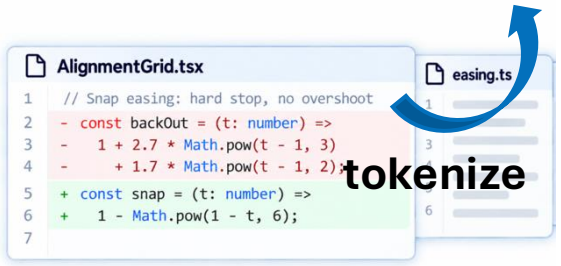
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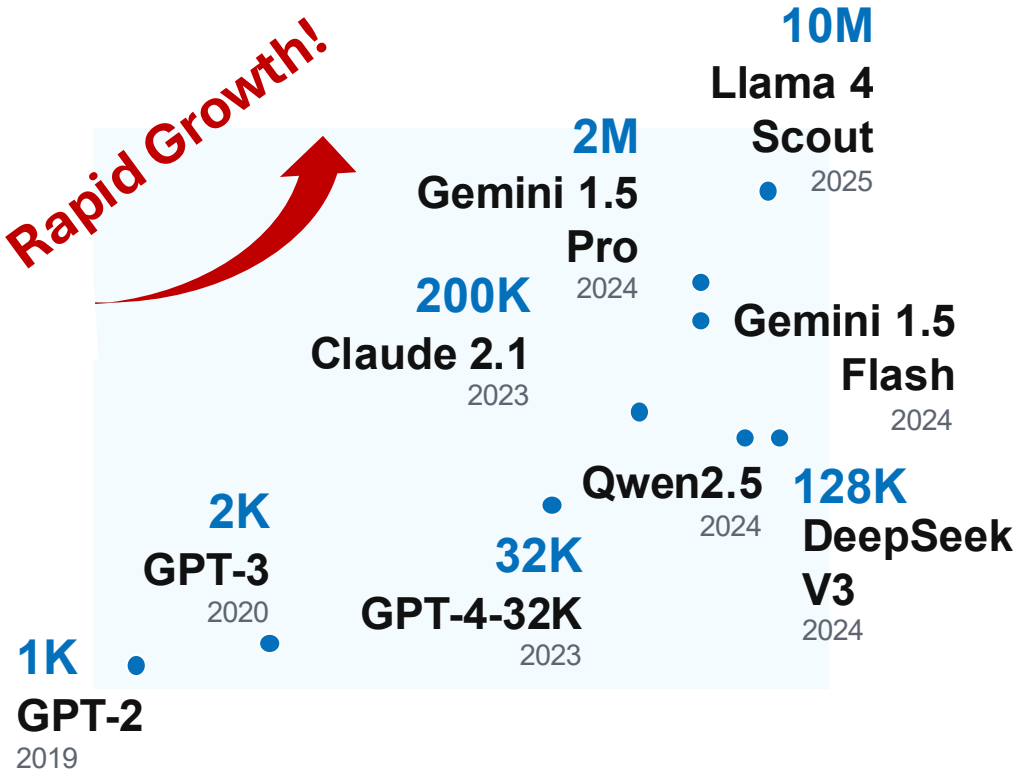
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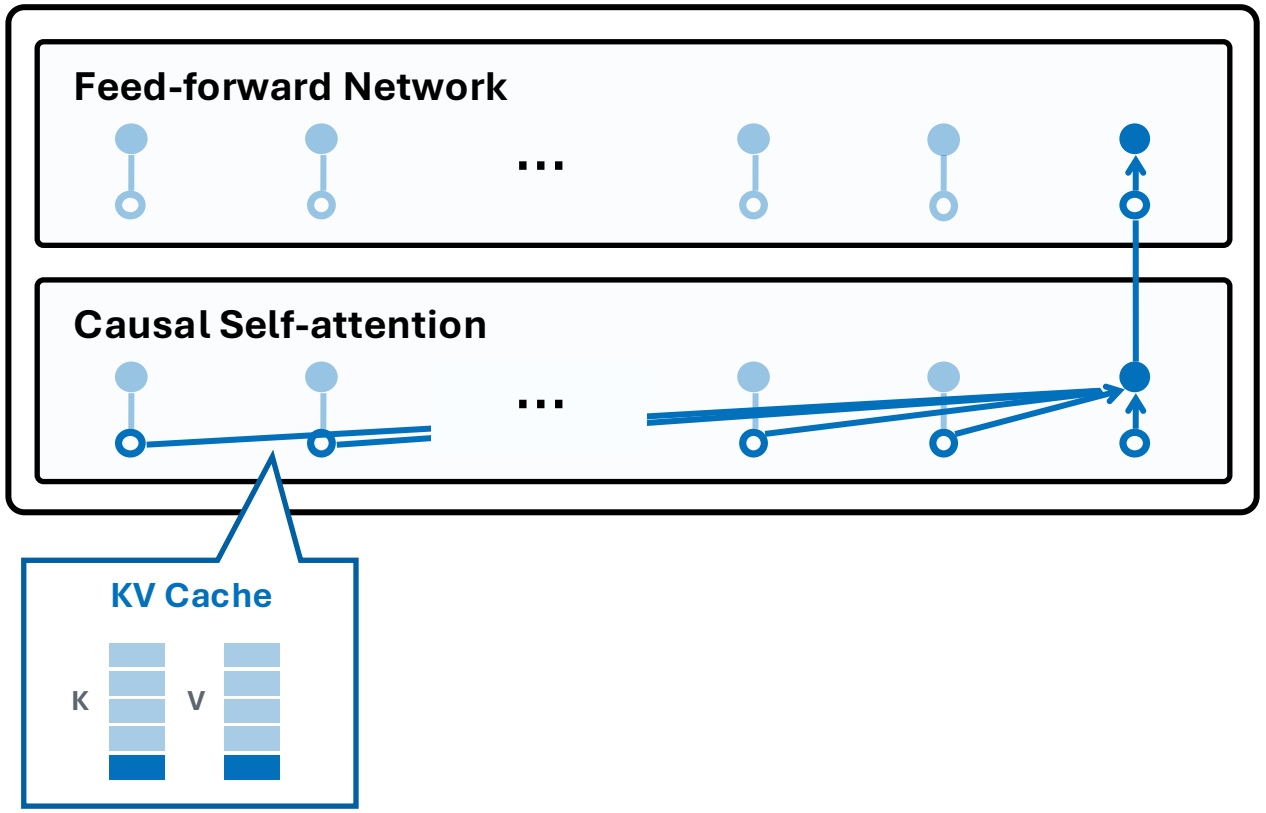


More tokens for harder tasks and longer interactions.



Growth in the token capacity of advanced LLMs over years.

However, longer context comes at costs …



With more context

KV-cache memory **O(n)**

Store more keys and values across layers.

Prefill attention compute **O(n²)**

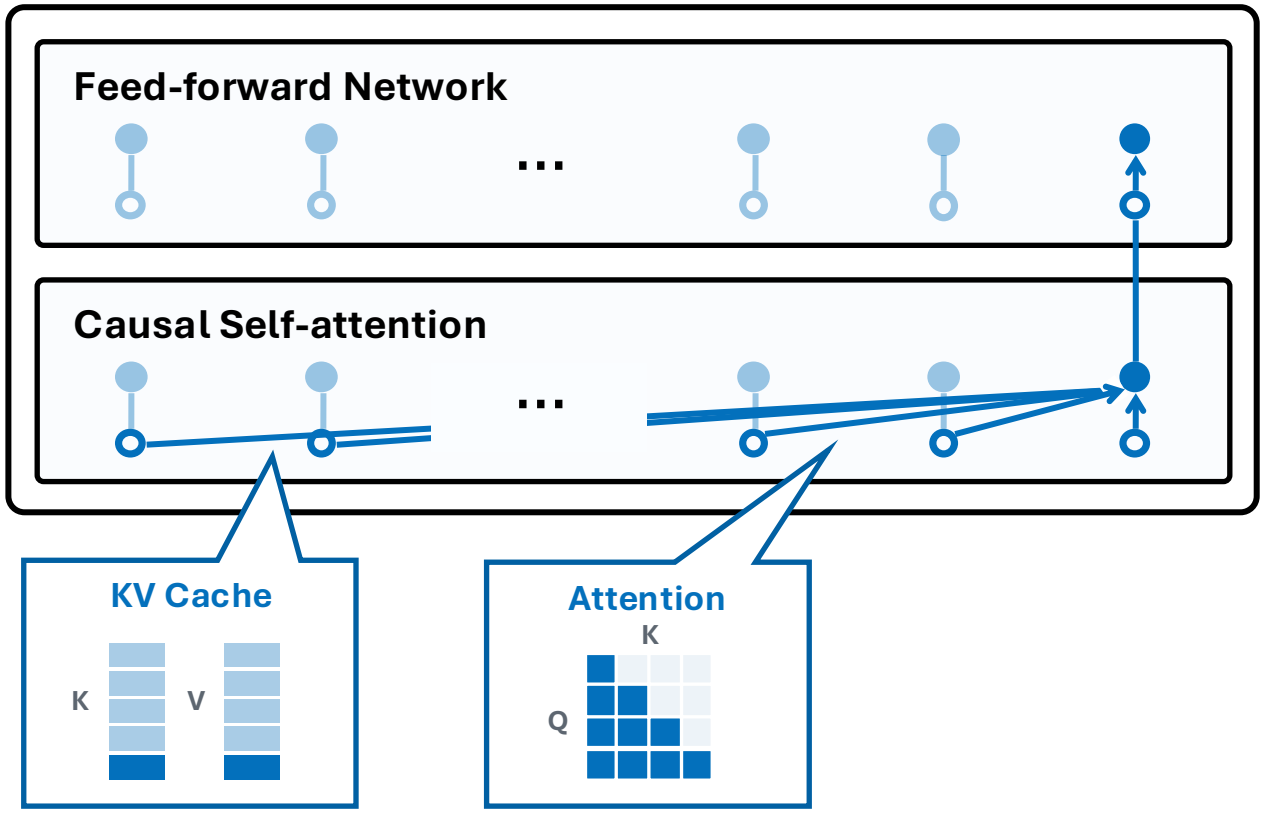
Process more prompt-token pairs with dense causal attentions.

Lost in the middle

Relevant information can be harder to use when it appears in the middle.

Longer contexts increase **memory demands** and **computational costs**, while **effective use of relevant information** remains challenging.

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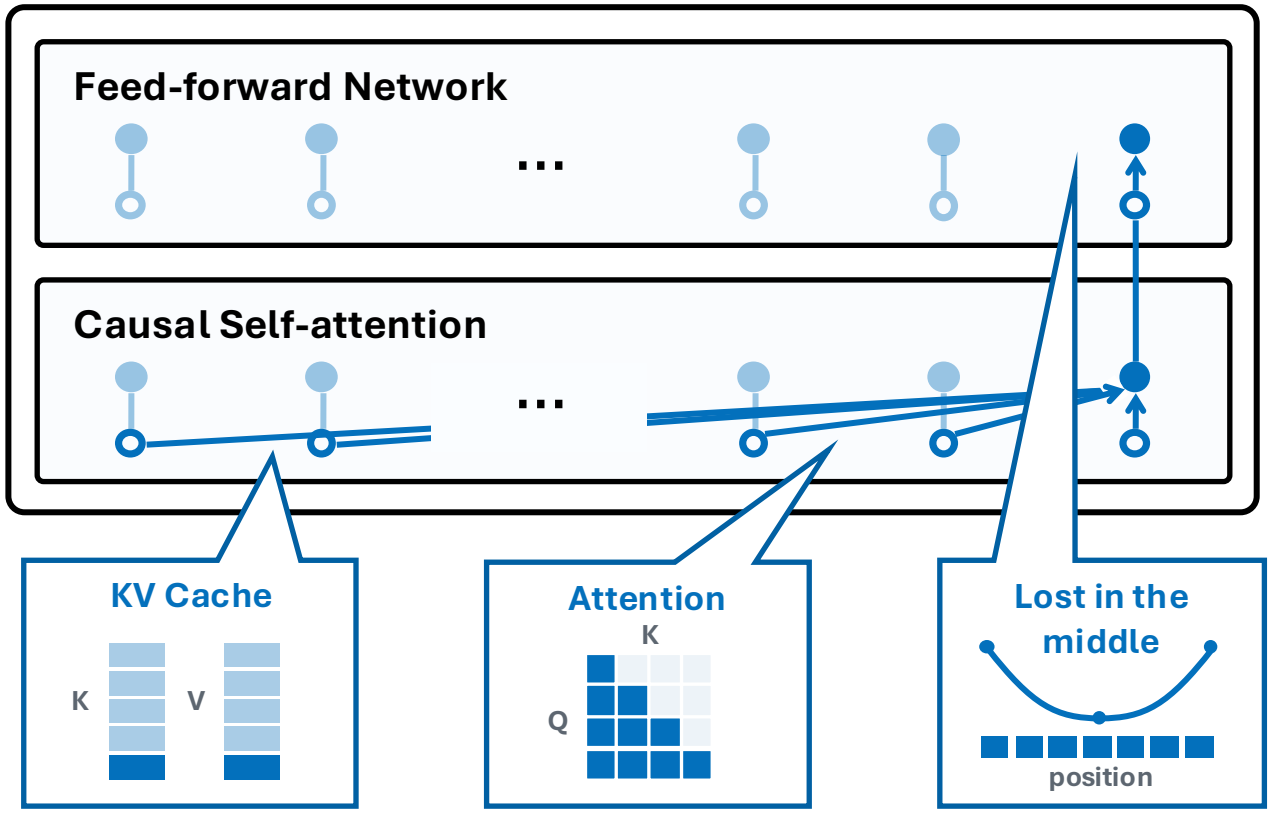
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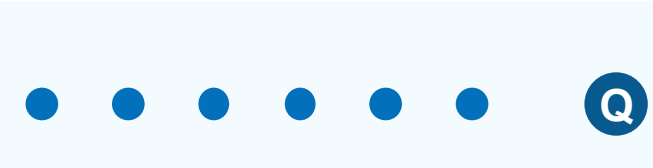
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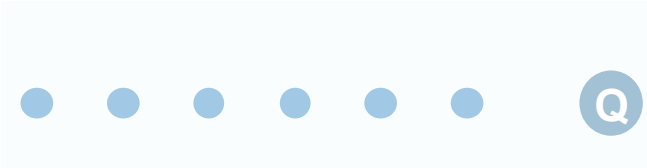
Sparse attention

Attention to a selected subset of the most important tokens

a) Local / global attention

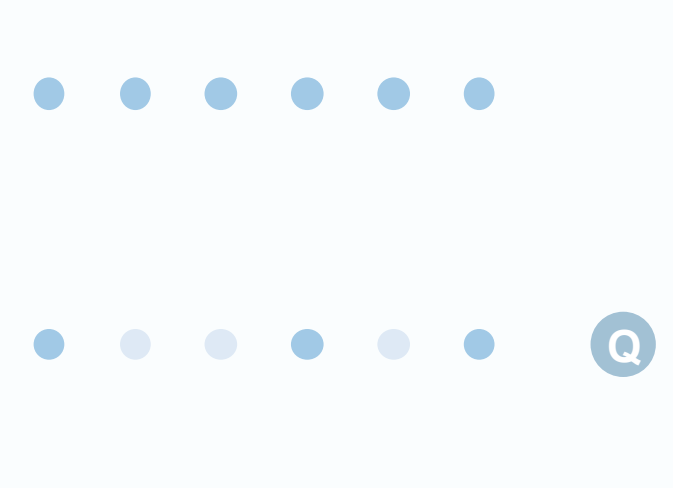


b) Query-dependent top-k



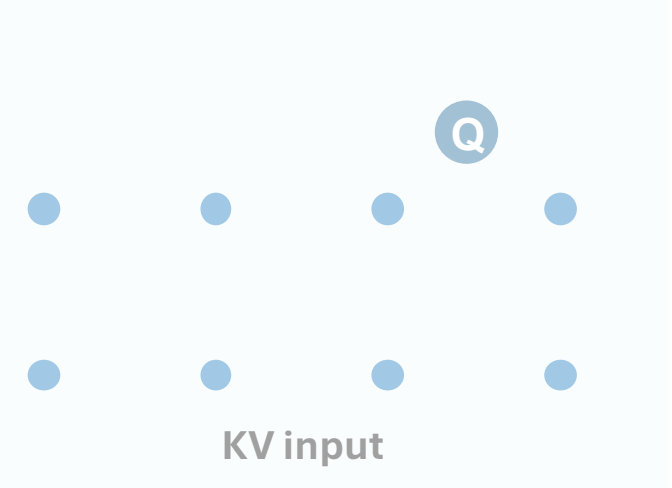
KV-cache reduction

Reduce memory by evicting useless KV entities



Linear attention

Aggregate past KV pairs into a fixed size recurrent state



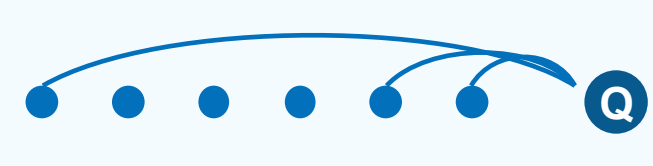
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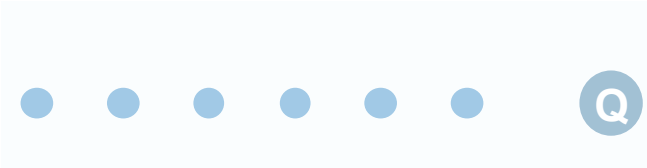
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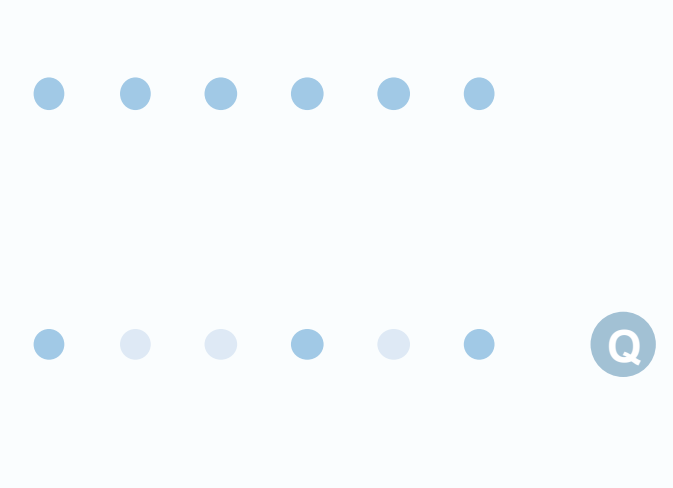


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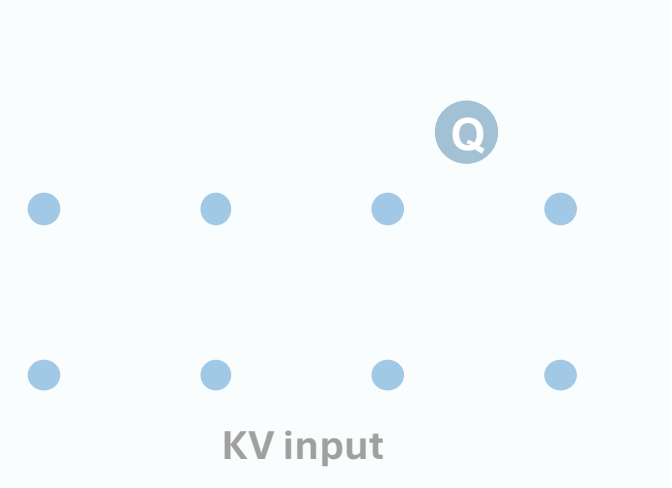
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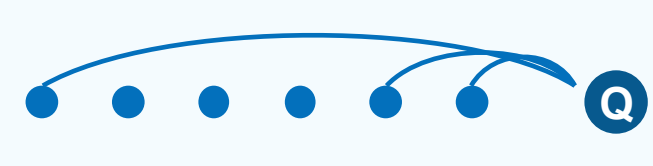
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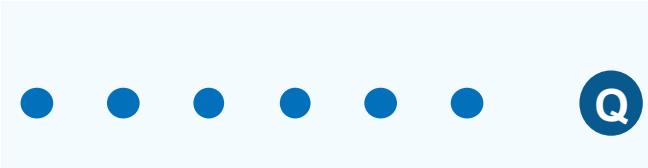
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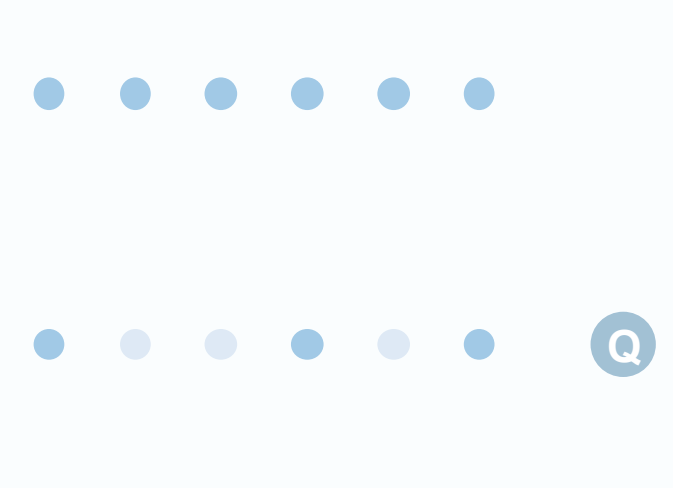


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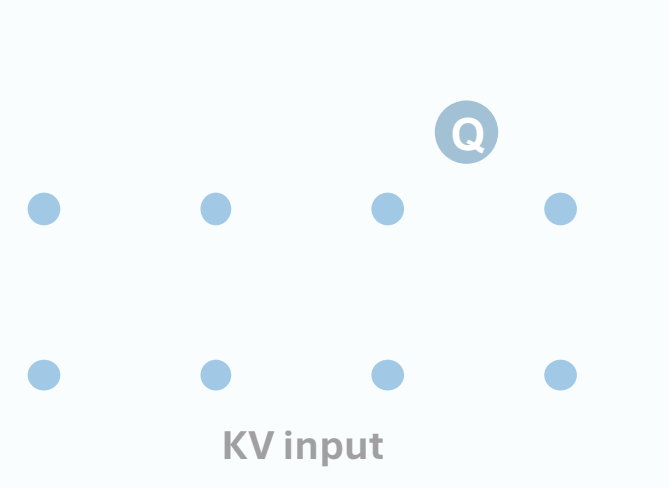
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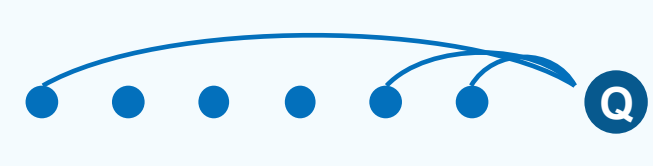
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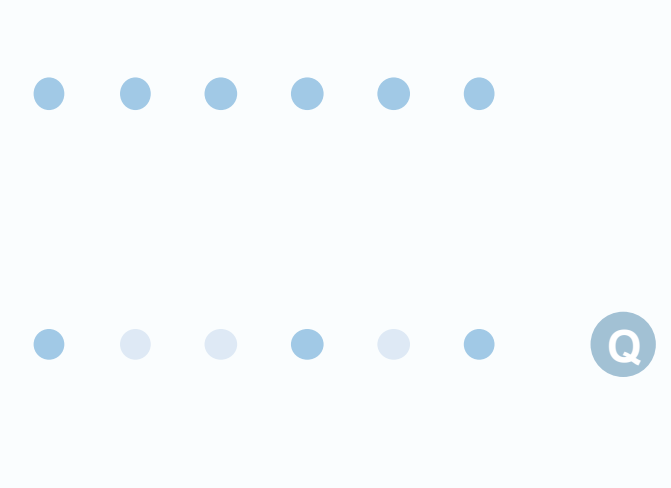


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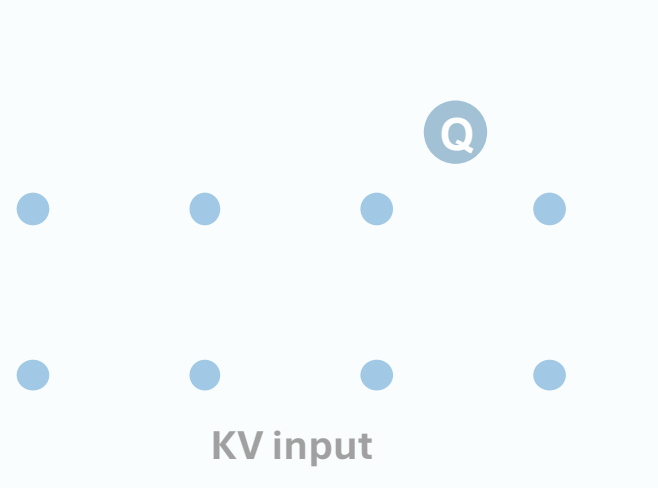
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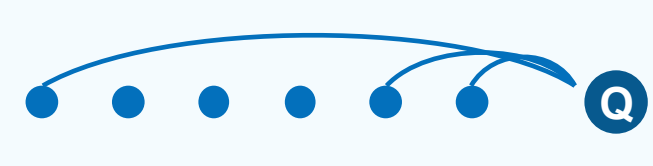
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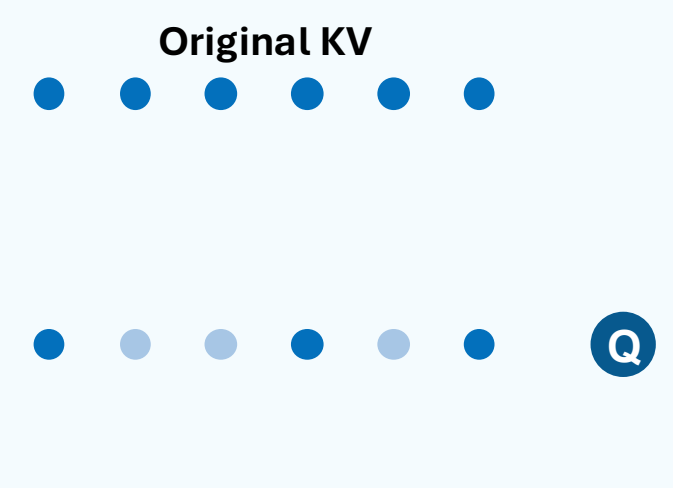


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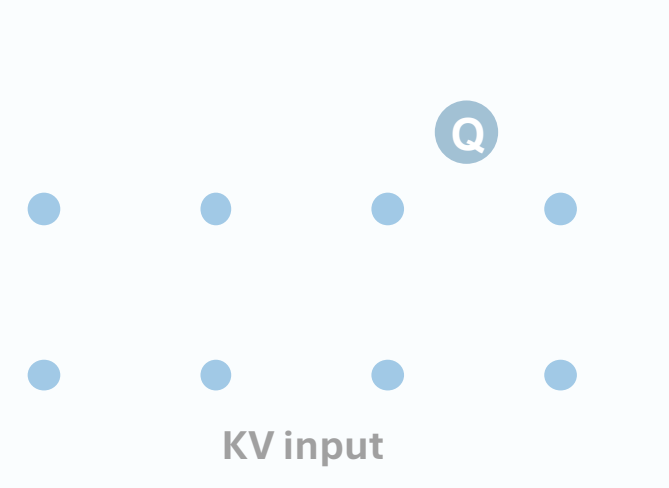
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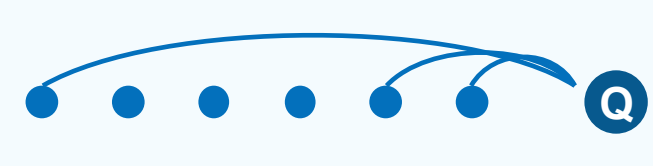
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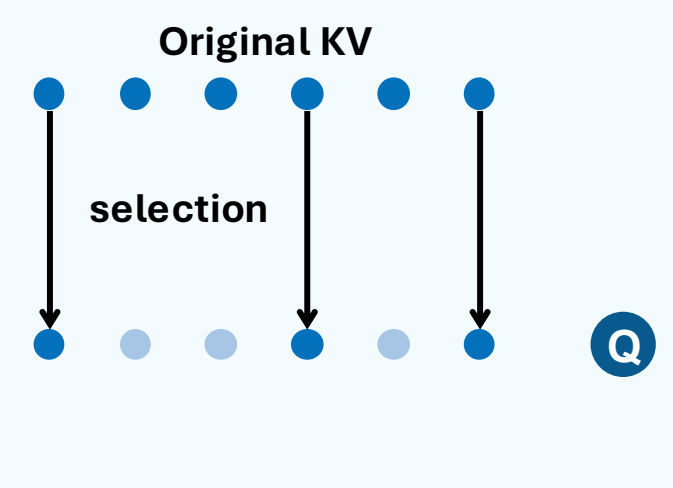


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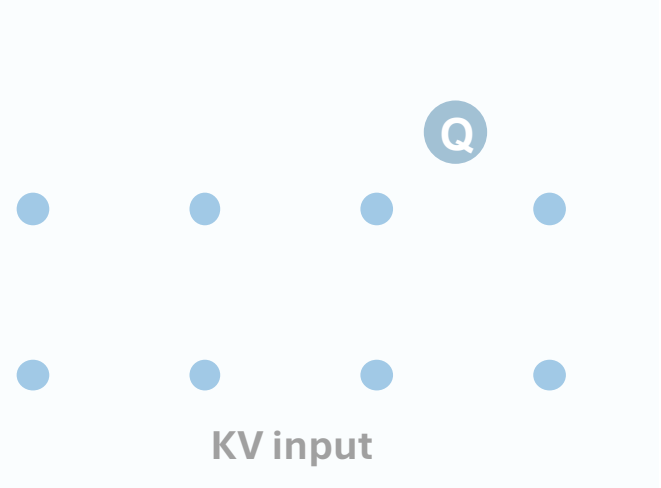
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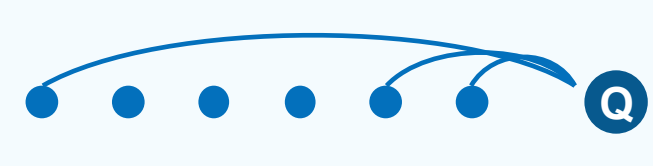
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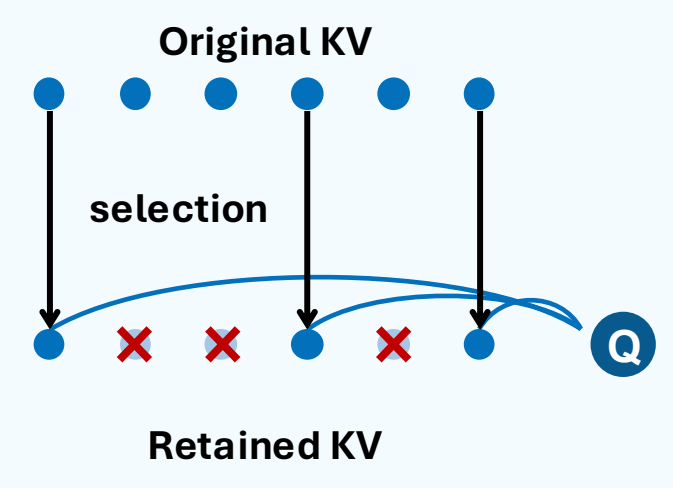


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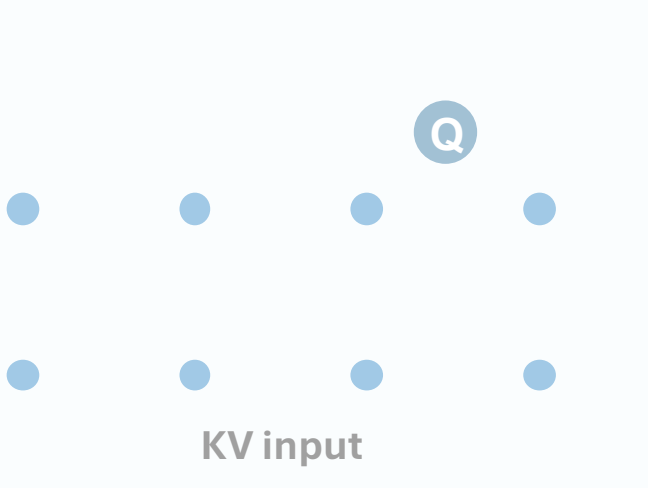
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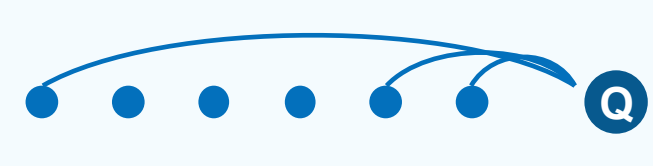
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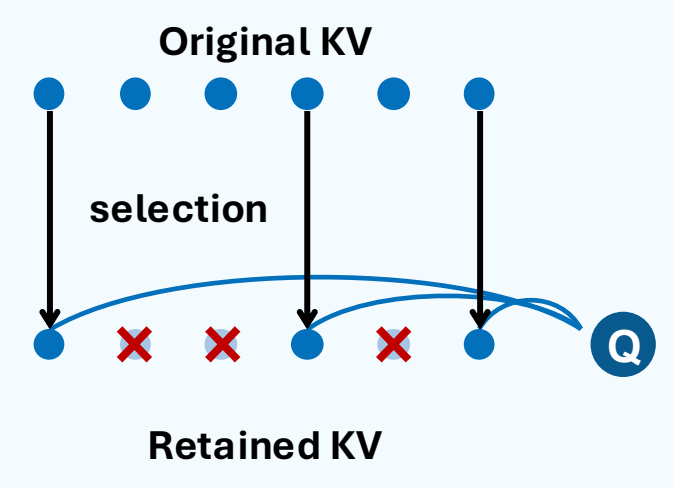


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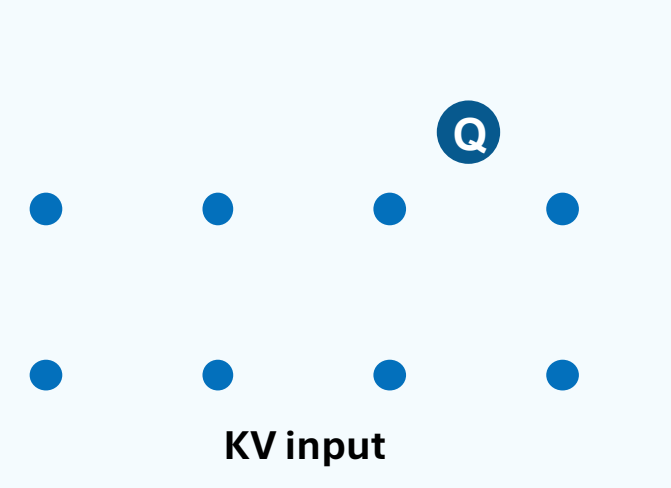
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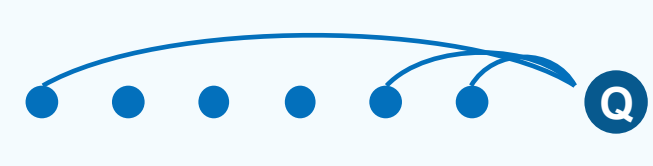
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Prior work changes how attention is computed

Sparse attention

Attention to a selected subset of the most important tokens

a) Local / global attention

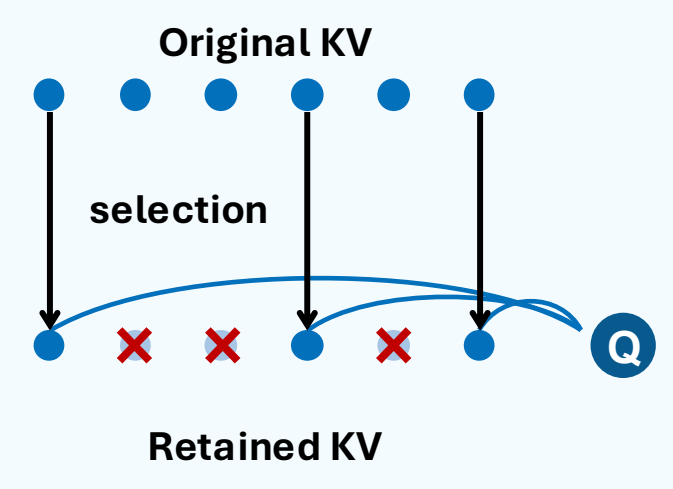


b) Query-dependent top-k



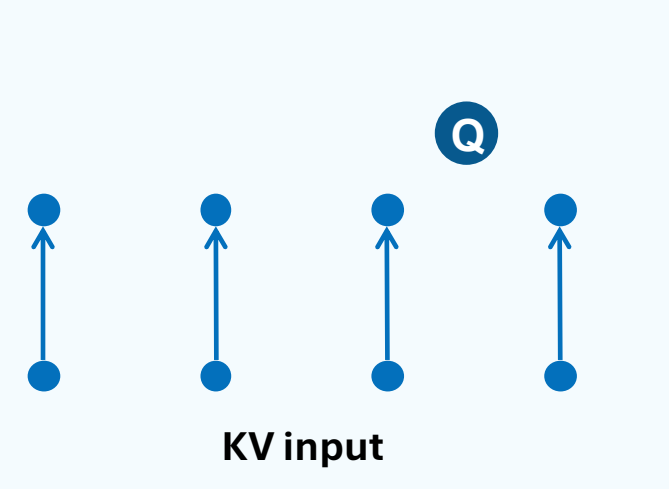
KV-cache reduction

Reduce memory by evicting useless KV entities



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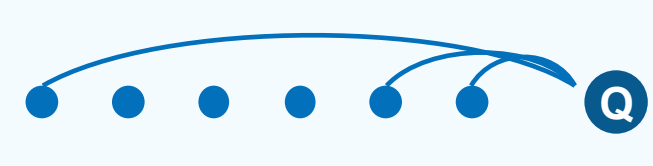
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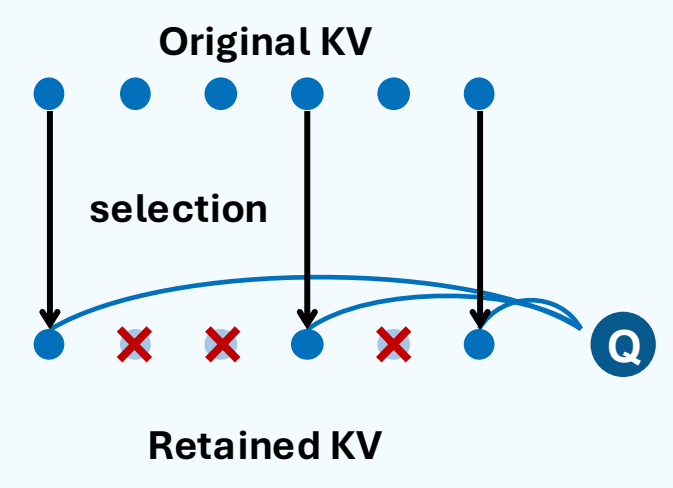


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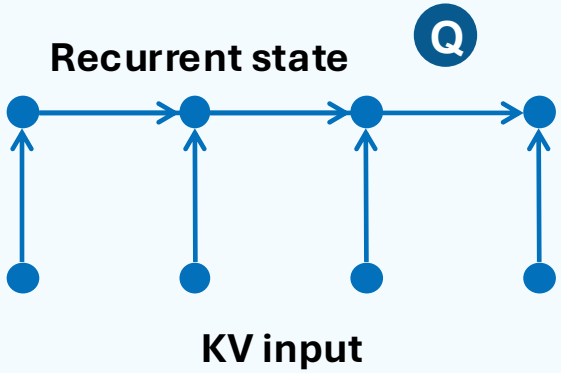
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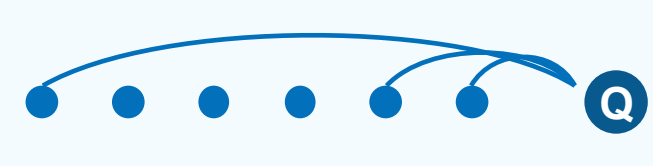
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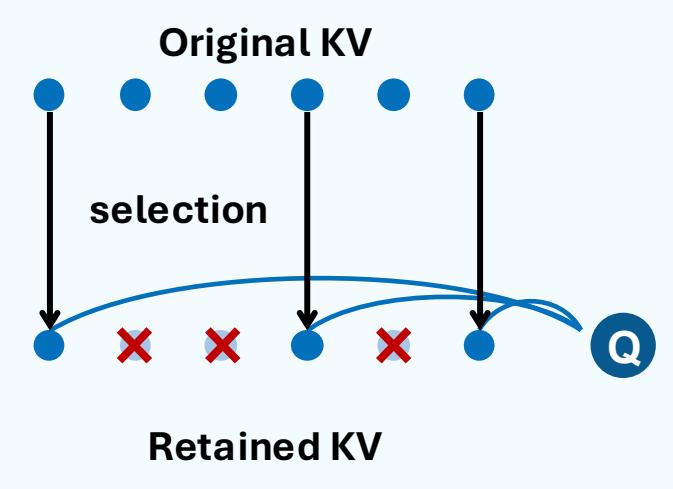


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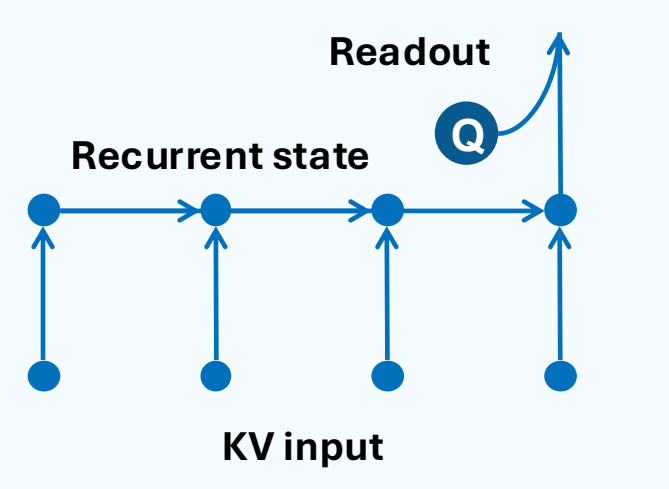
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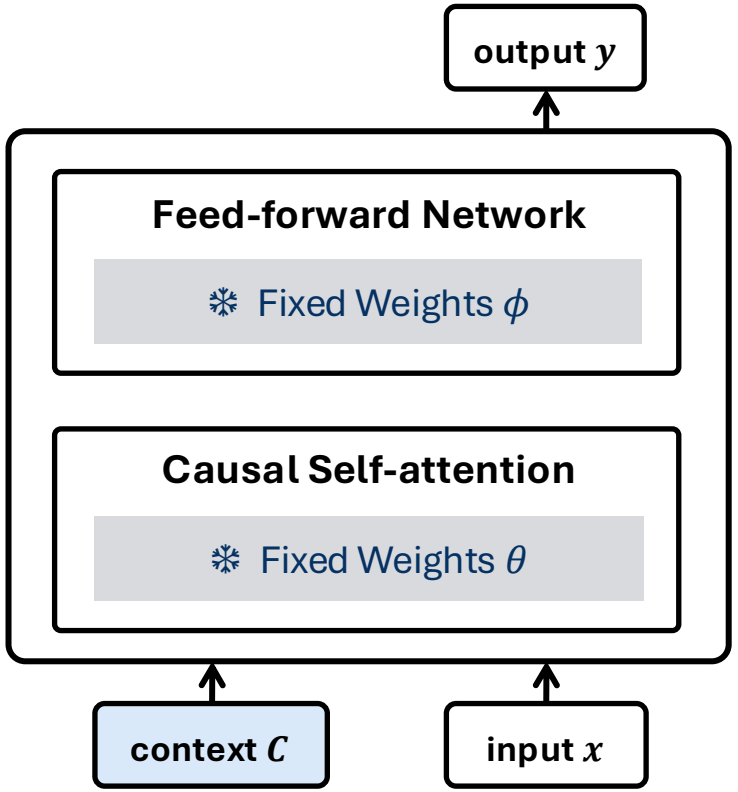


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Test-time training as an alternative recipe

Test-time context encoding

encode context into model parameters and remove it from the input.



Our Goal $p_{\theta_C}(y \mid x) \approx p_{\theta_0}(y \mid C, x)$

Preserve the effect of context on predictions, without retaining it in the input.

Objective. $L(\phi; C) = -\log p(C; \phi)$

Standard autoregressive loss, updating only the fast weights

Why it is promising

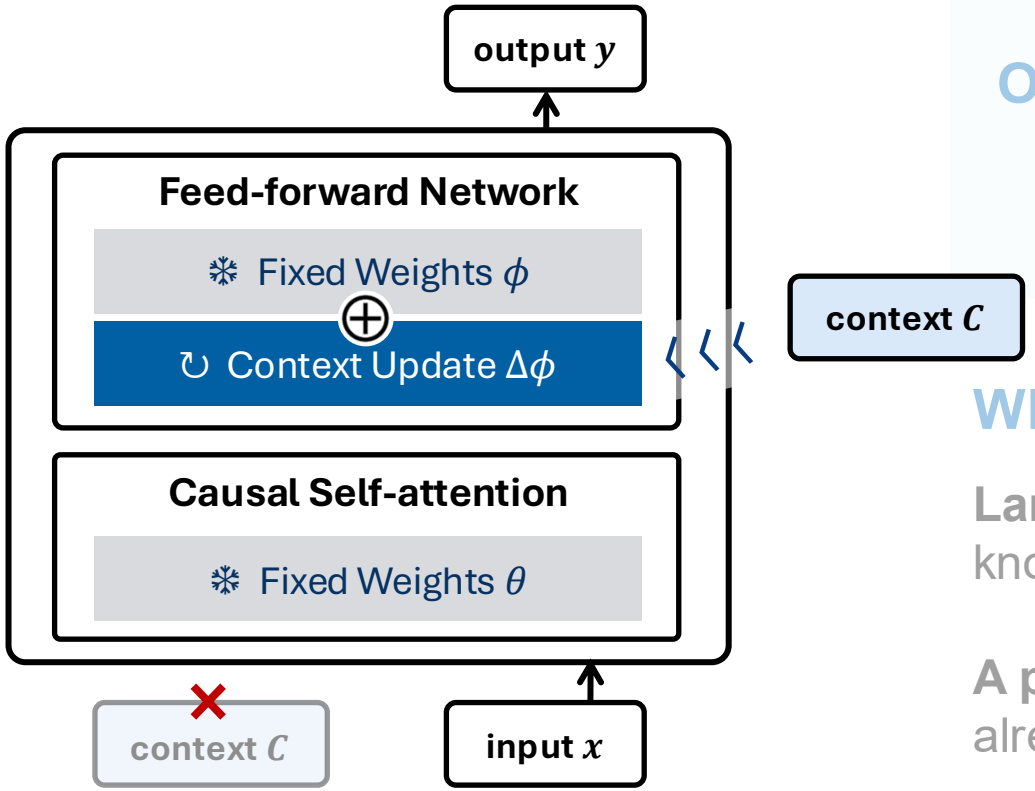
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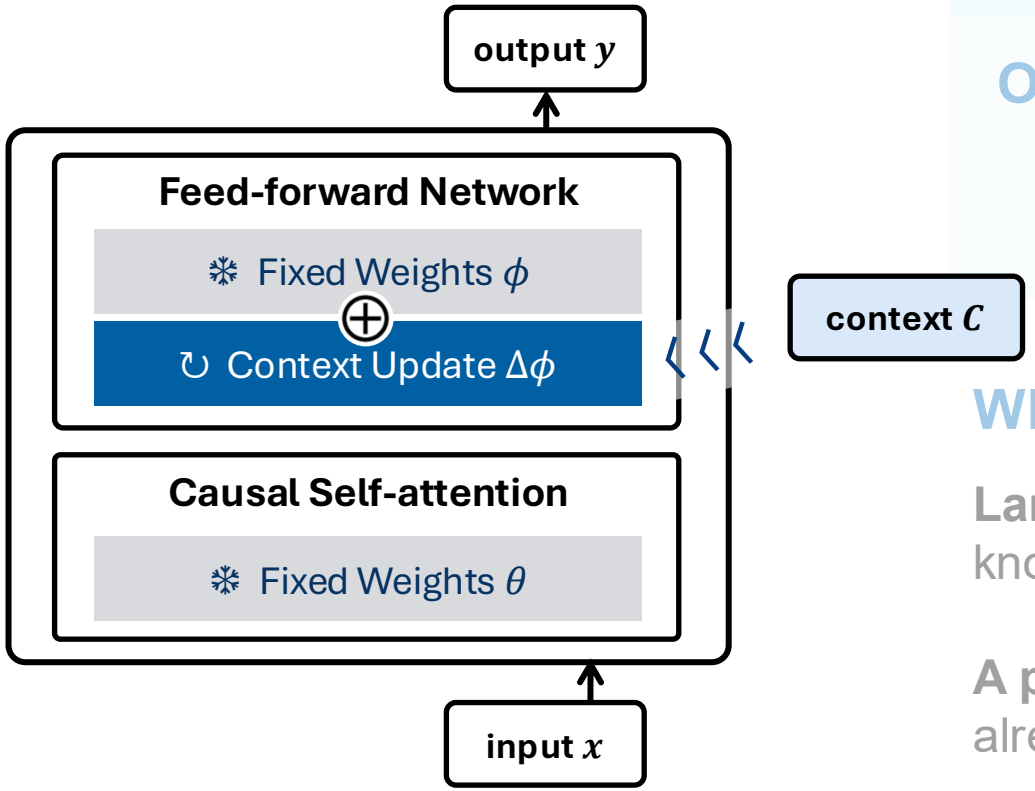
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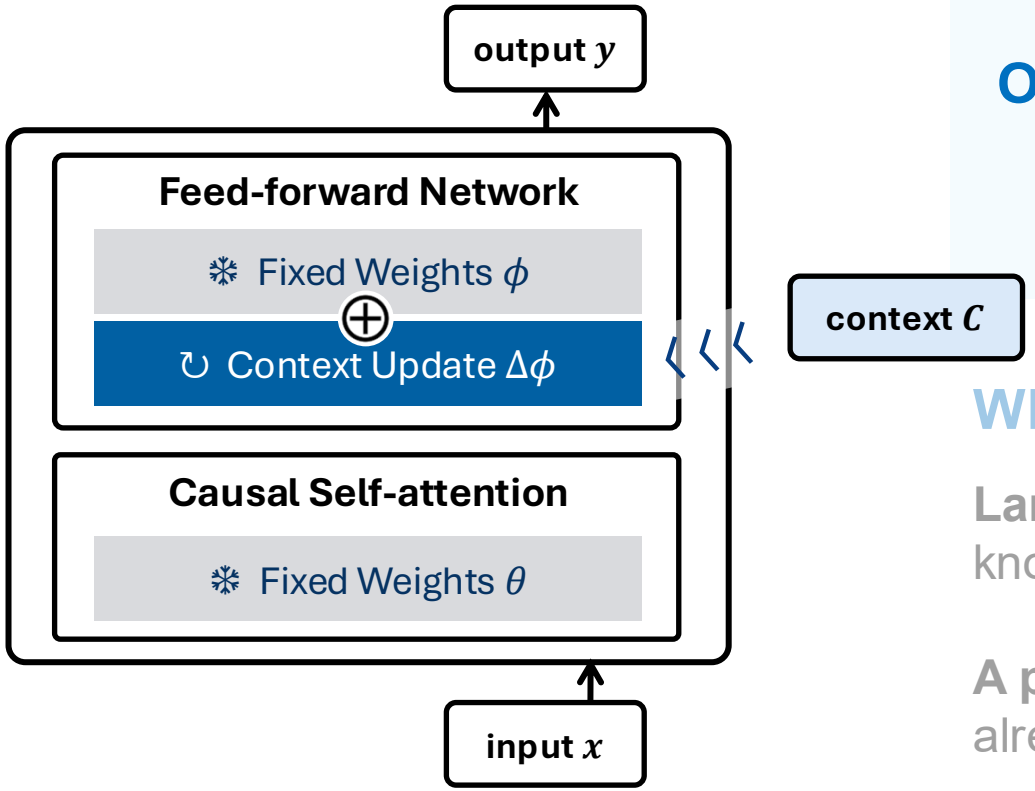
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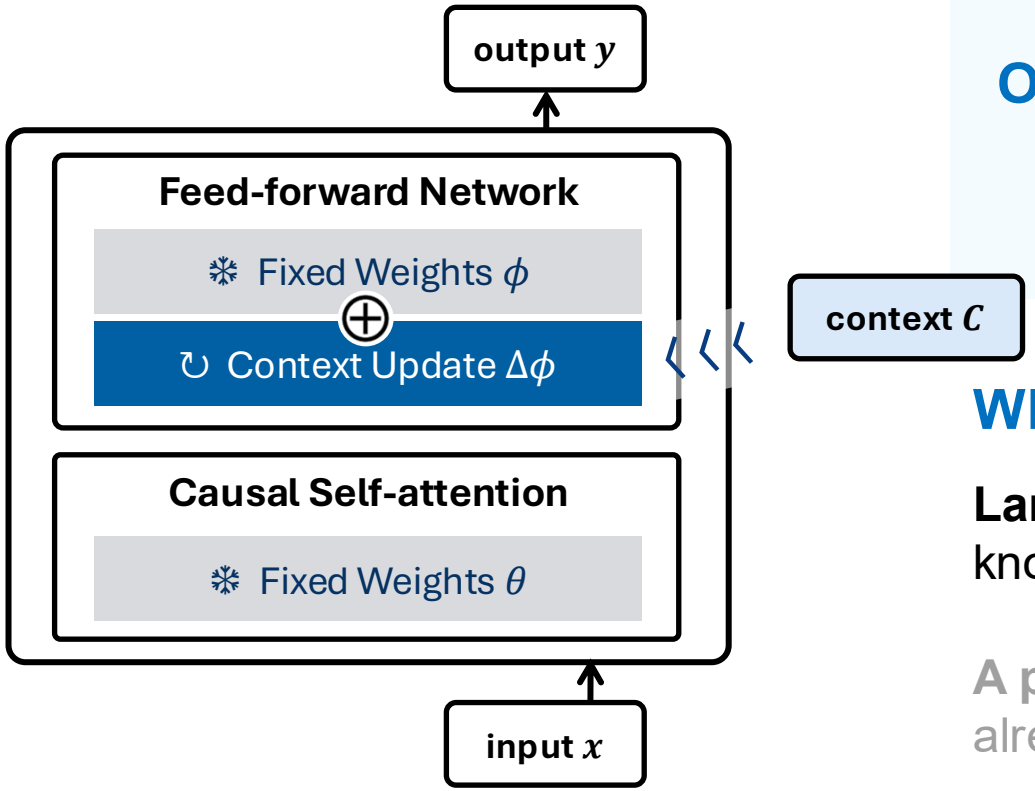
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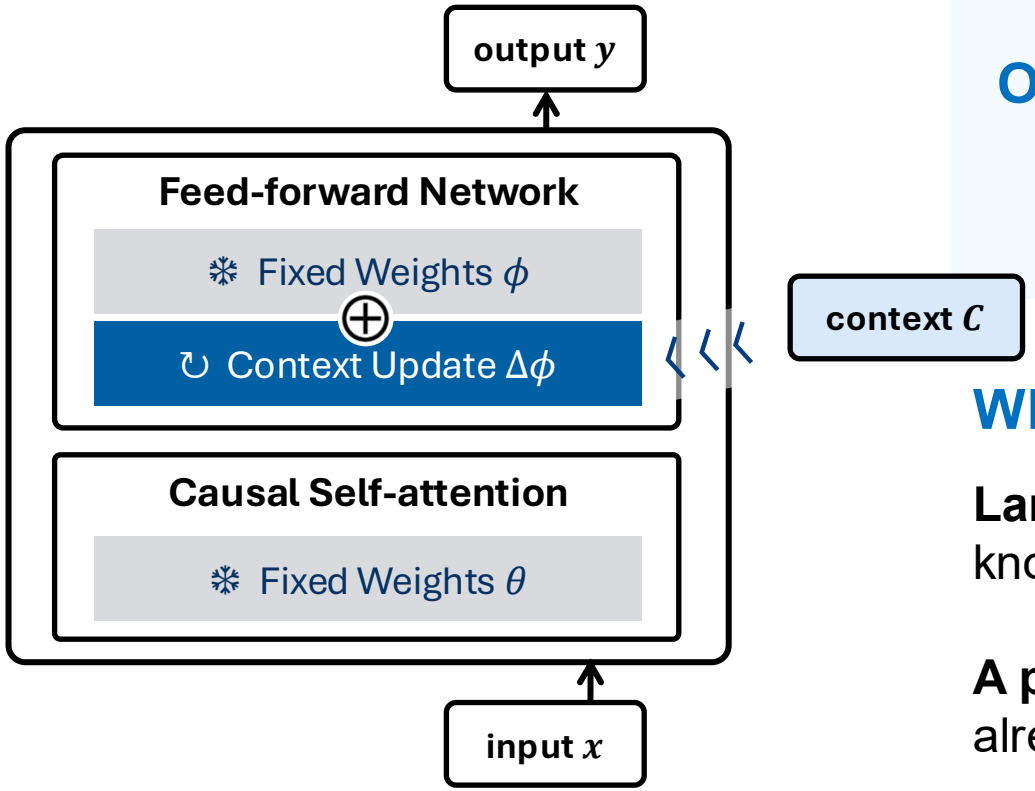
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Supporting and Opposing Views



Low cost: Updating a small subset of parameter is enough for context encoding.

System changes: Supporting TTT requires redesigning the entire parallel serving infrastructure.





How LLMs already work: LLMs learn knowledge through autoregressive training and recall it in response to input.

Unpredictable behavior: How LLMs use learned knowledge remains implicit and difficult to control.





Favor recent knowledge: Models tend to favor information recently encoded through gradient updates.

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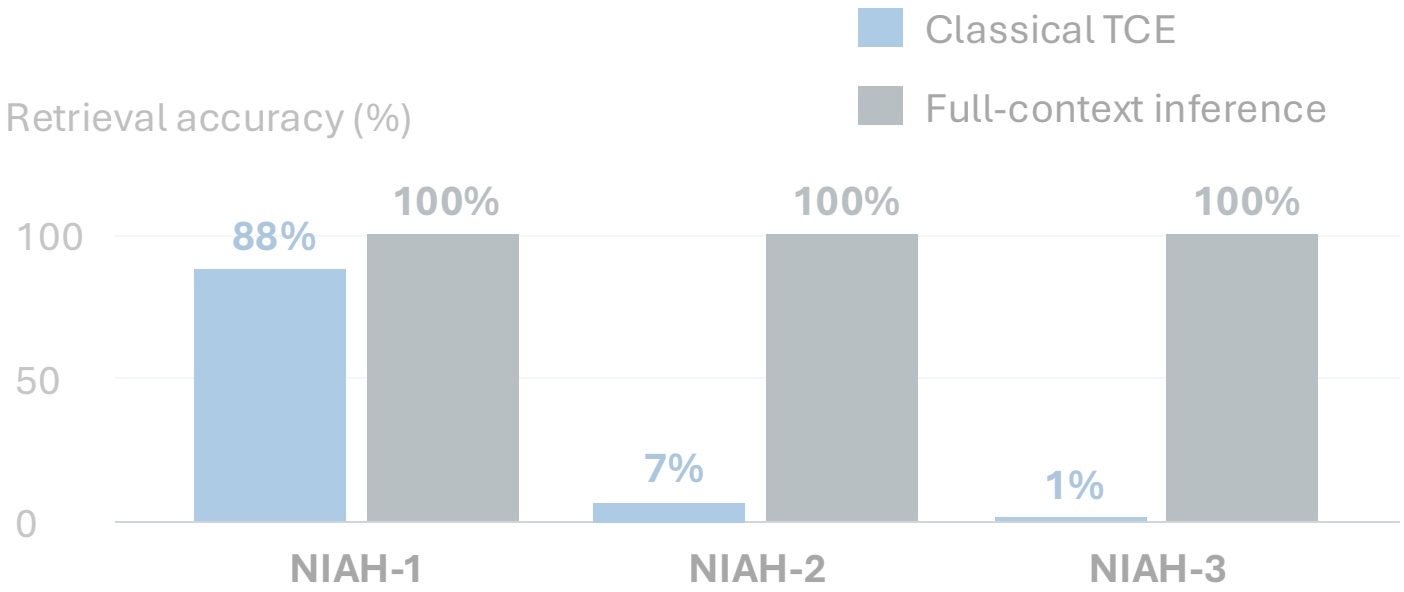
Needle in a haystack

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z1ds 0kia mjkt 8pkw q300 qip8 vnda 2qw6 p13j vh2
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k1pl xbuk q6h9 sjfb clny p665 hwqz hhj1 ktol 4j9
xvu3 pqdy s1e9 1tkv bxhn 7tdq ct81 oc0x 2tfa p2x

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Qwen3.5-9B



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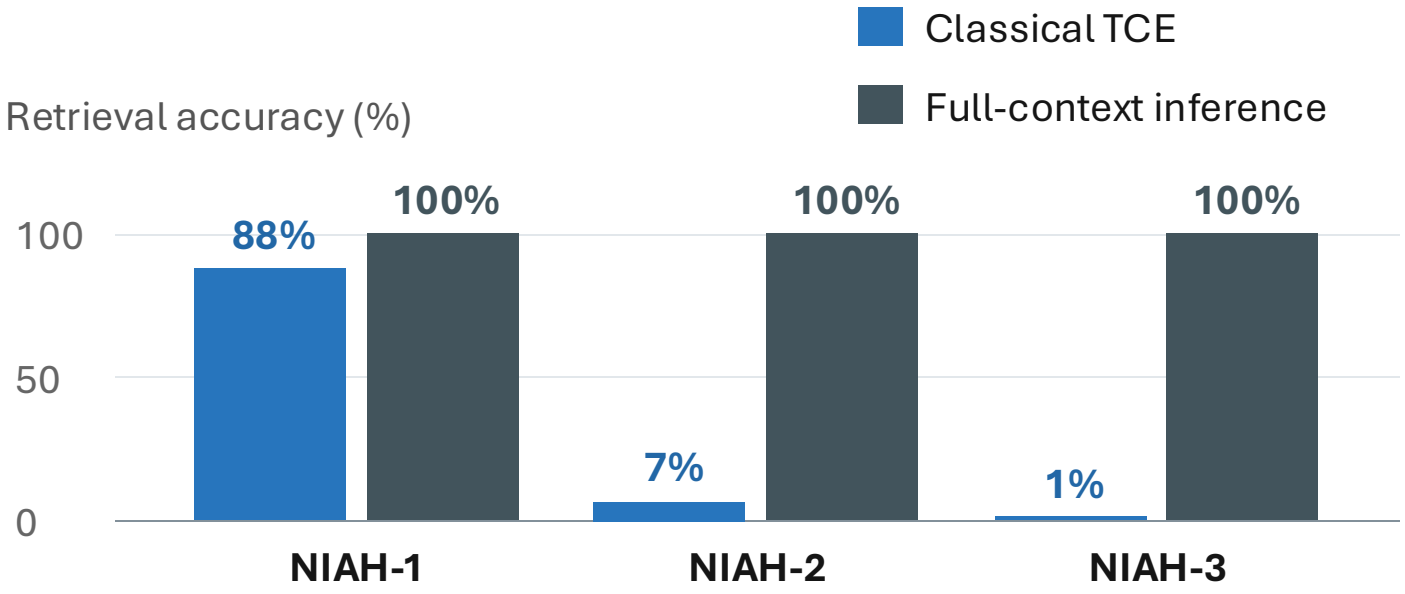
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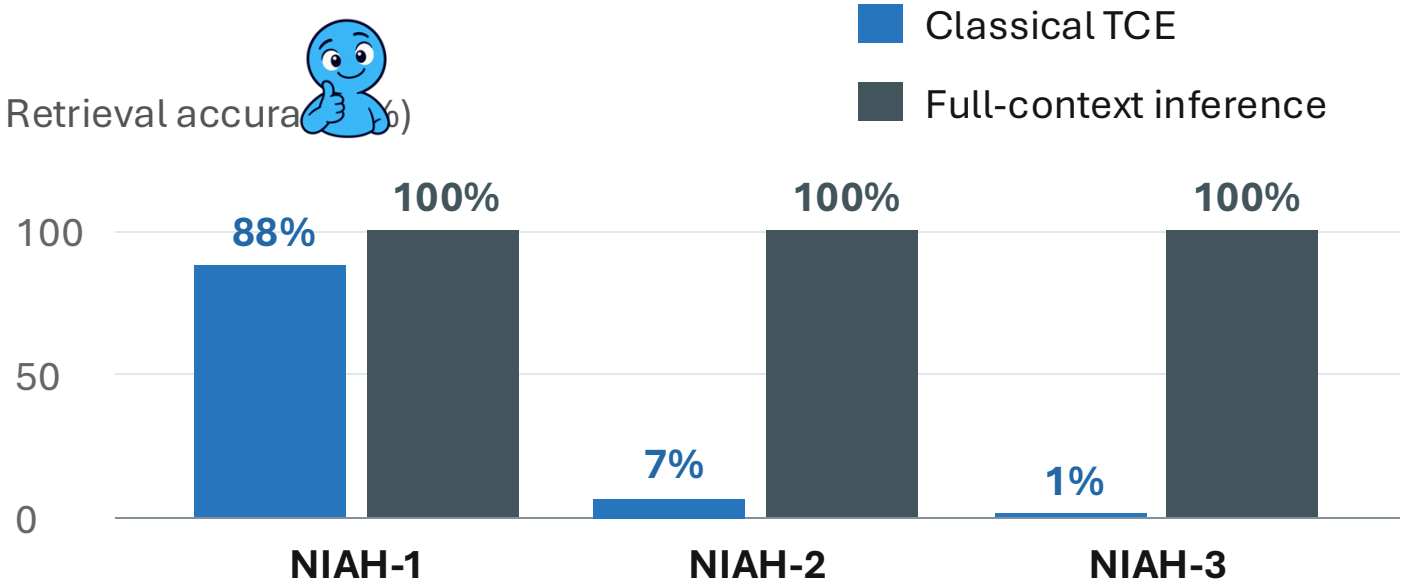
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z2t9 58xp ystq bfo8 by09 m64s xlhd qnnj rele xd8
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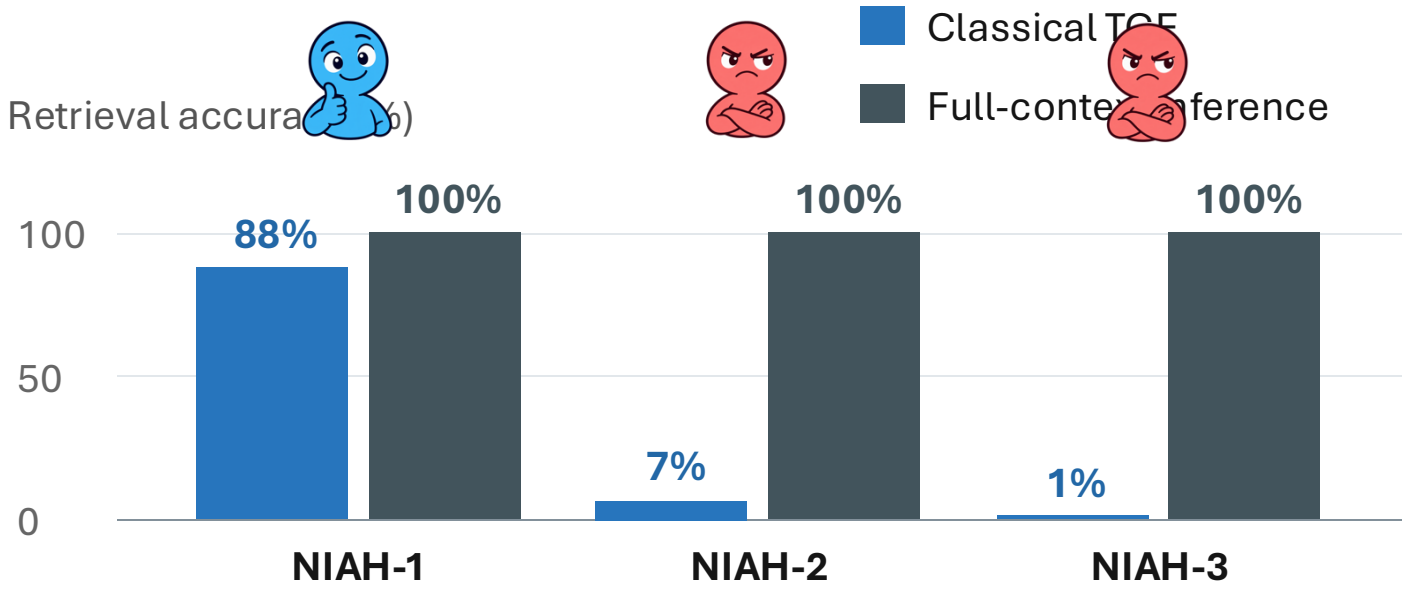
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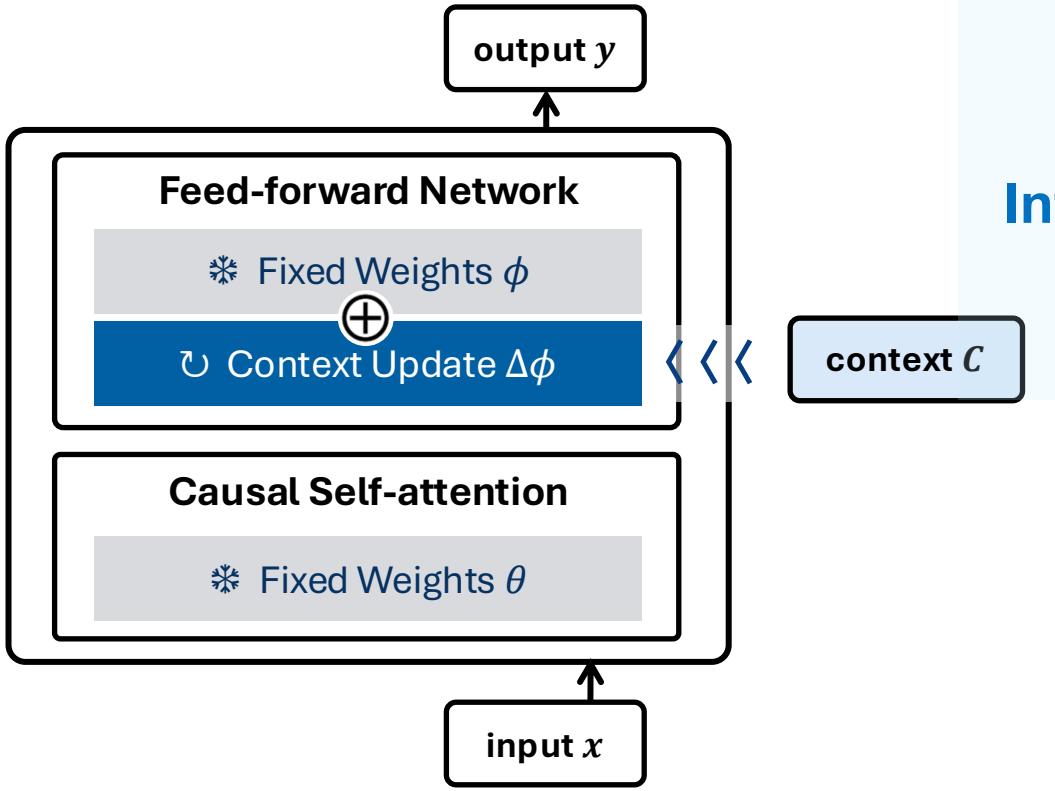
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How can we control context encoding?

Original Test-time context encoding

encode context into model parameters and remove it from the input.



Encoding $\phi_C \leftarrow \text{argmax} \log p(C; \phi)$

Encoding context into fast weights via autoregressive loss

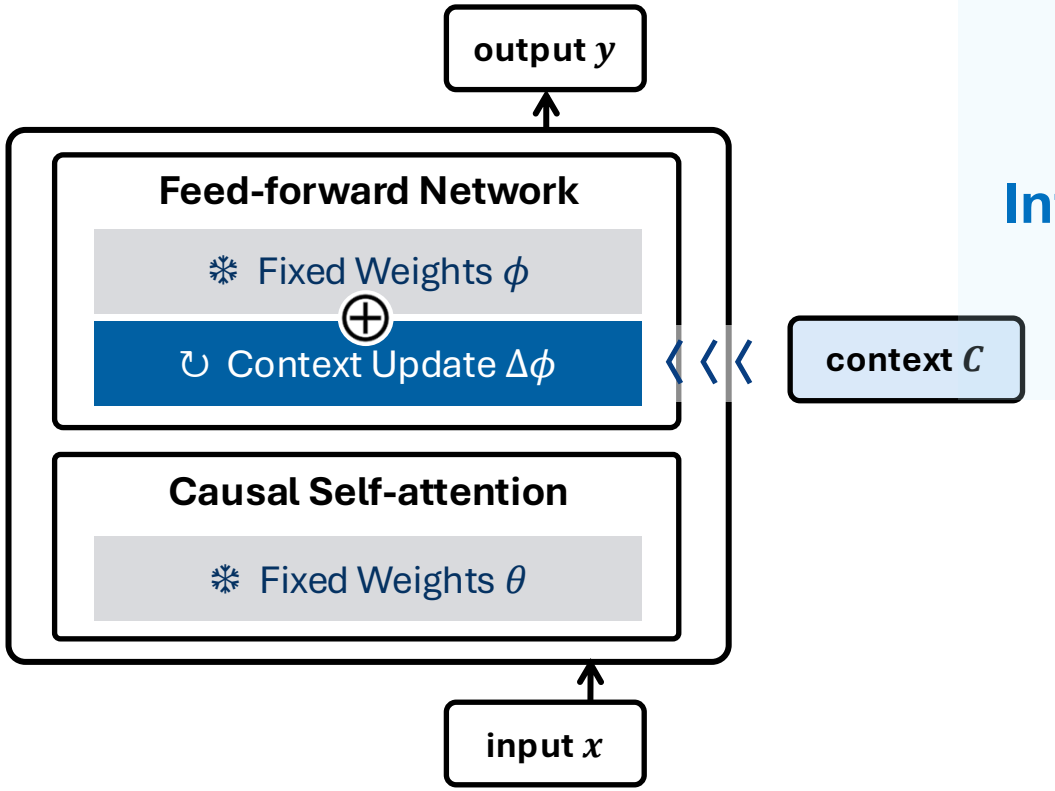
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The updated model performs inference without the original context as input.

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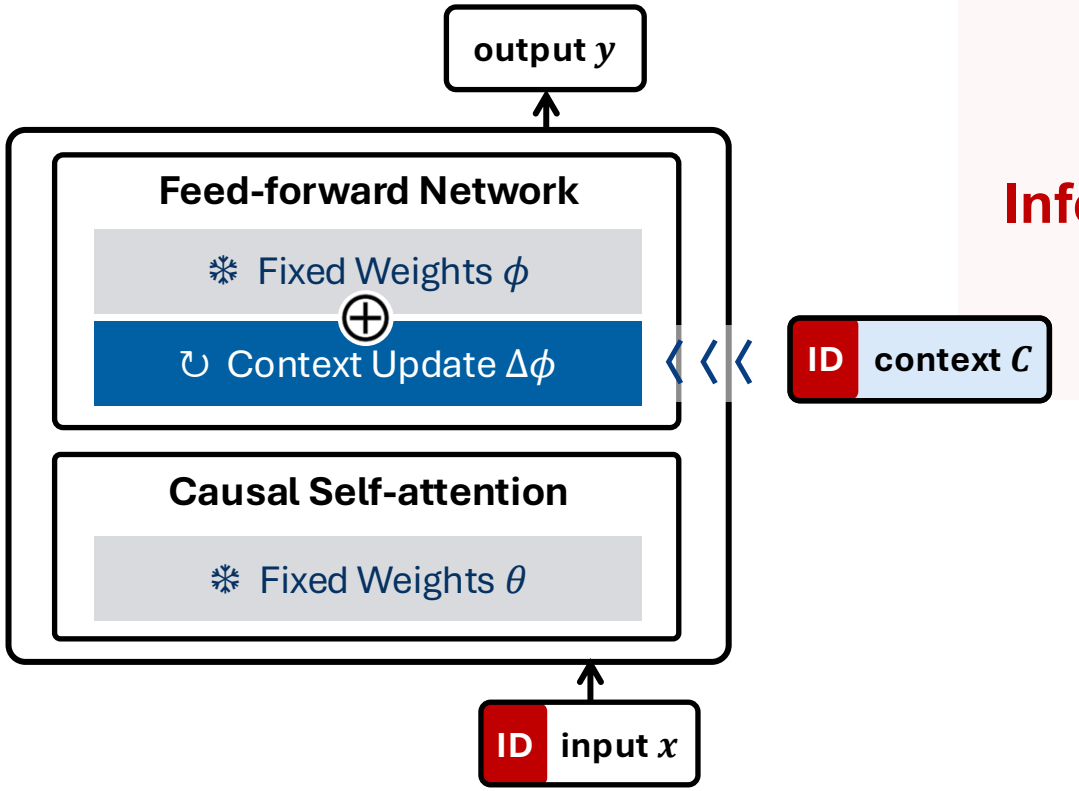
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We lack explicit control over model behavior after context encoding!

How can we control context encoding?

Addressable context encoding (ACE)

assign unique ID to each context during test-time training and inference.



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Encoding context conditioned on the UID into the fast weights

Inference $y \sim p(\cdot \mid uid, x; \phi_C)$

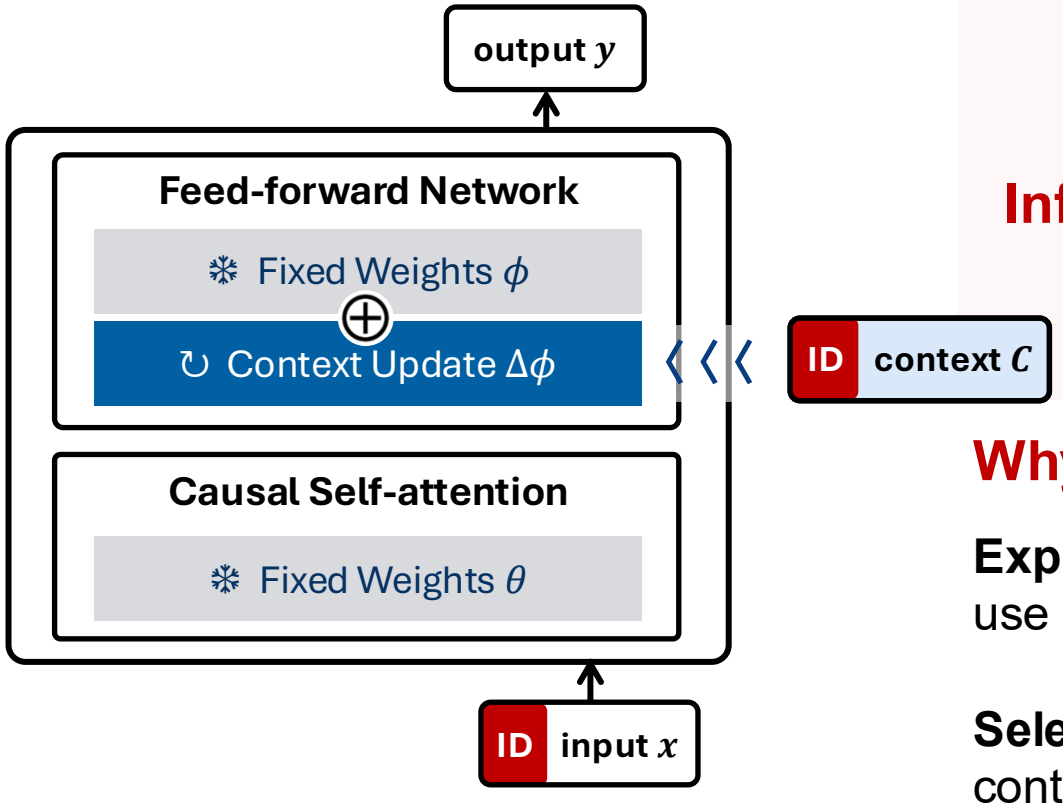
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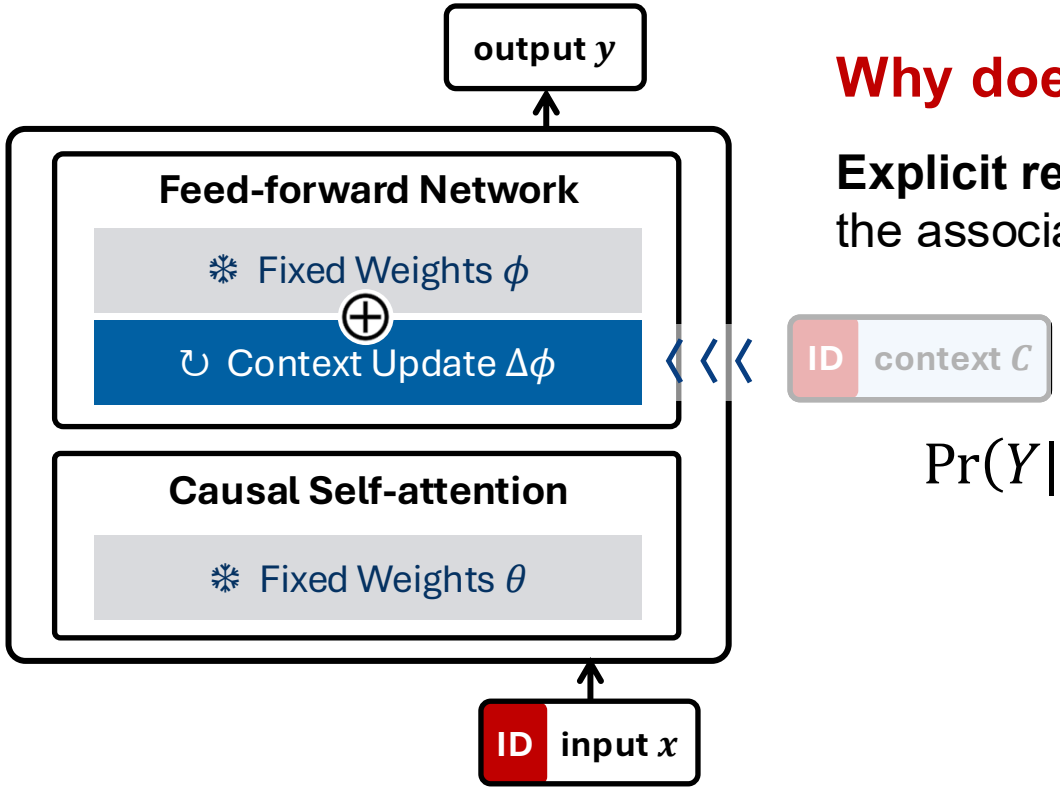
Explicit retrieval cue. The UID encourages the model to use the associated context during inference.

Selective context recall. Unique IDs distinguish encoded contexts and let us specify which one to recall.

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increase

$$\Pr(Y|uid, x) \approx \Pr(Y|R = C, x)\Pr(R = C|uid) + \Pr(Y|R \neq C, x)\Pr(R \neq C|uid)$$

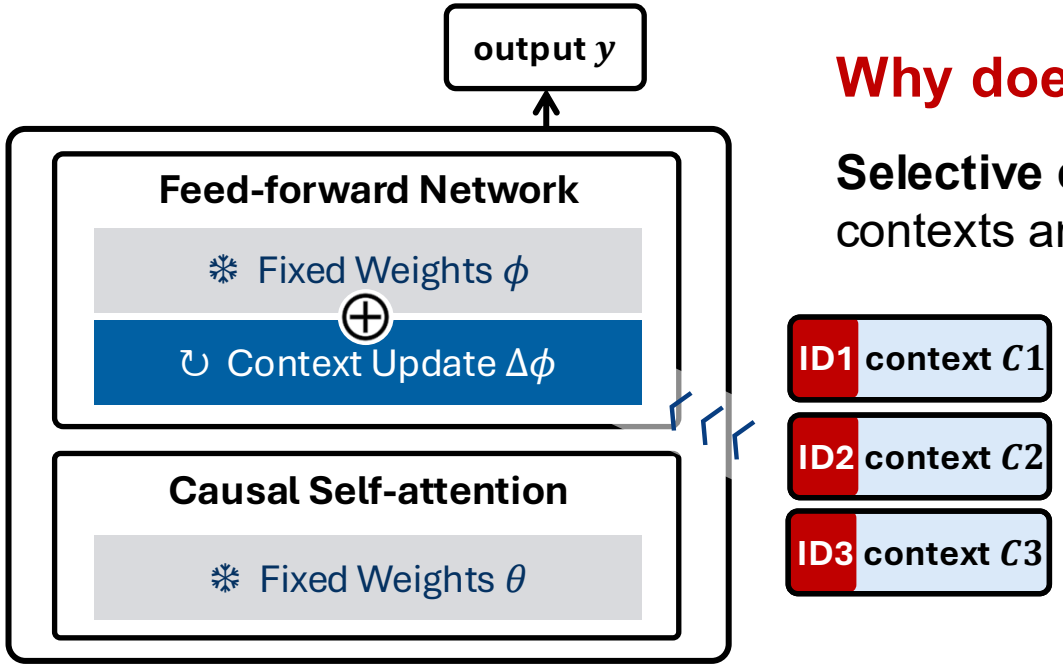
decrease

assume $Pr(R = C|uid, x) = Pr(R = C|uid)$
& $Pr(Y|R, x) = (Y|R, uid, x)$

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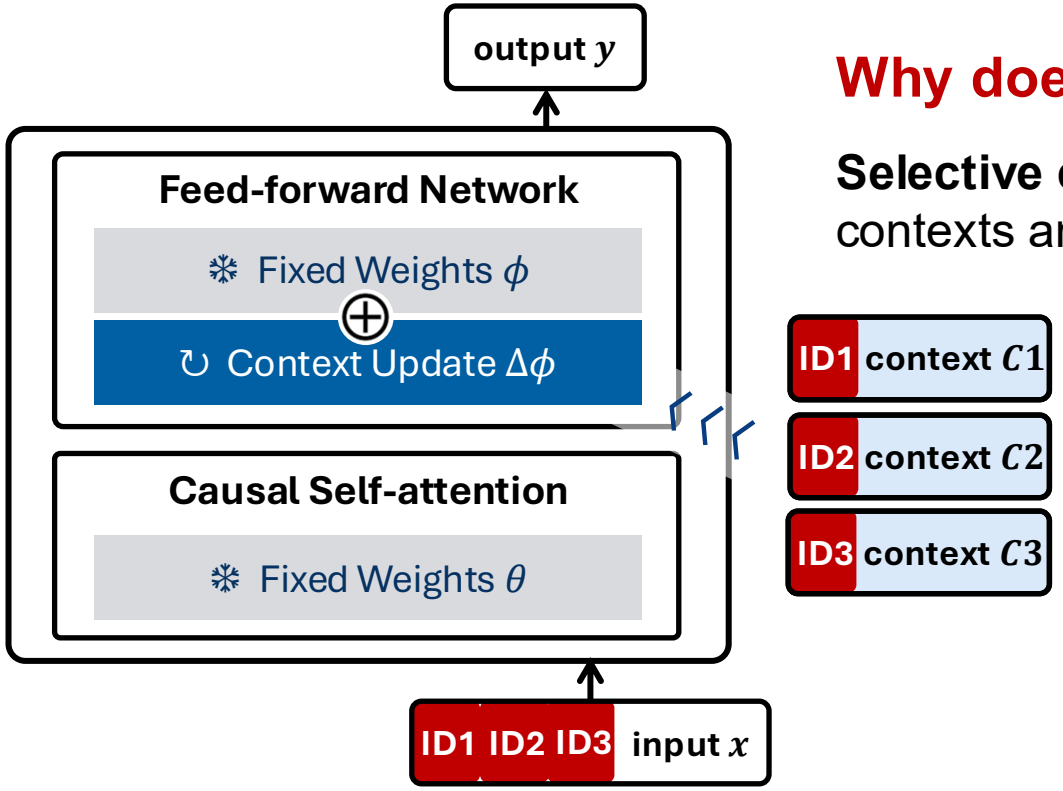
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When encoding context sequentially, UIDs allow us to

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Why does it work?

Selective context recall. Unique IDs distinguish encoded contexts and let us specify which one to recall.

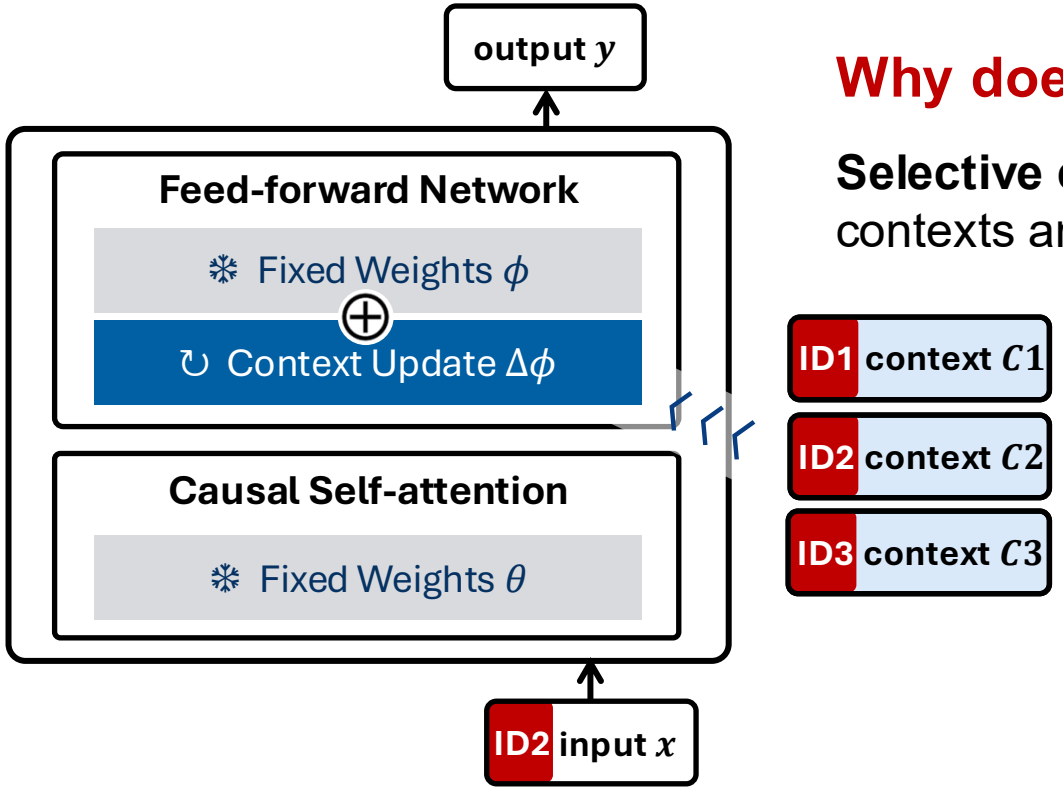
When encoding context sequentially, UIDs allow us to

explicitly reference each context during inference;

How can we control context encoding?

Addressable context encoding (ACE)

Encoding $\phi_C \leftarrow \text{argmax} \log p(C|uid; \phi)$ **Inference** $y \sim p(\cdot \mid uid, x; \phi_C)$



Why does it work?

Selective context recall. Unique IDs distinguish encoded contexts and let us specify which one to recall.

When encoding context sequentially, UIDs allow us to

explicitly reference each context during inference;

selectively use which contexts the model should use.

How can we control context encoding?

Prompt Enhancement ACE

Encoding $\phi_C \leftarrow \text{argmax} \log p(C|uid;\phi)$ **Inference** $y \sim p(\cdot \mid uid, x; \phi_C)$

Role-binding prompt

```
The following text is a context chunk from the current conversation at s_t, identified
by i_t. It has been encoded into the model and should be treated as conversational
context whenever it is invoked: c_t
```

This text is part of our conversation.
Remember it under this UID.

Context-invocation prompt

```
Parts of the conversation have been encoded a                    time:
i_1 at s_1, ..., i_T at s_T. Treat these chunks as prior conversational context in
subsequent inference. Selectively use information relevant to the current request
without reproducing unrelated details.
```

How can we control context encoding?

Prompt Enhancement ACE

Encoding $\phi_C \leftarrow \text{argmax} \log p(C|uid;\phi)$ **Inference** $y \sim p(\cdot \mid uid, x; \phi_C)$

Role-binding prompt

```
The following text is a context chunk from the current conversation at s_t, identified
by i_t. It has been encoded into th                                  conversational
context whenever it is invoked: c_
```

Use the conversation stored under these UIDs to answer the current question. Ignore irrelevant details.

Context-invocation prompt

```
Parts of the conversation have been encoded as the context chunks, ordered by time:
i_1 at s_1, ..., i_T at s_T. Treat these chunks as prior conversational context in
subsequent inference. Selectively use information relevant to the current request
without reproducing unrelated details.
```

Controlled experiments



TCE ACE

Use encoded information

Recall

Wharf code is GC-483.

What is the code?

→ **GC-483**

4.6%
38.1%

Recognition

Wharf code is GC-483.

What were you just told?

→ **Wharf code is GC-483.**

2%
94.4%

Continuation

What is Norway's capital?

Continue the conversation.

→ **Oslo**

1.2%
5.4%

Follow context over prior knowledge

Counterfactual

Peru's capital is Arequipa.

What is Peru's capital?

→ **Arequipa**

14%
95%

Dispute

Phone inventor: A. Meucci

Who invented the phone?

→ **Antonio Meucci**

11.6%
97.7%

Outdated

Bayern's coach: V. Kompany

Who is the updated coach?

→ **Vincent Kompany**

10%
100%

~11-token contexts · Qwen3.5-9B

Needle-in-a-haystack

Find the exact information hidden in irrelevant text.

NIAH-1 Repetitive background

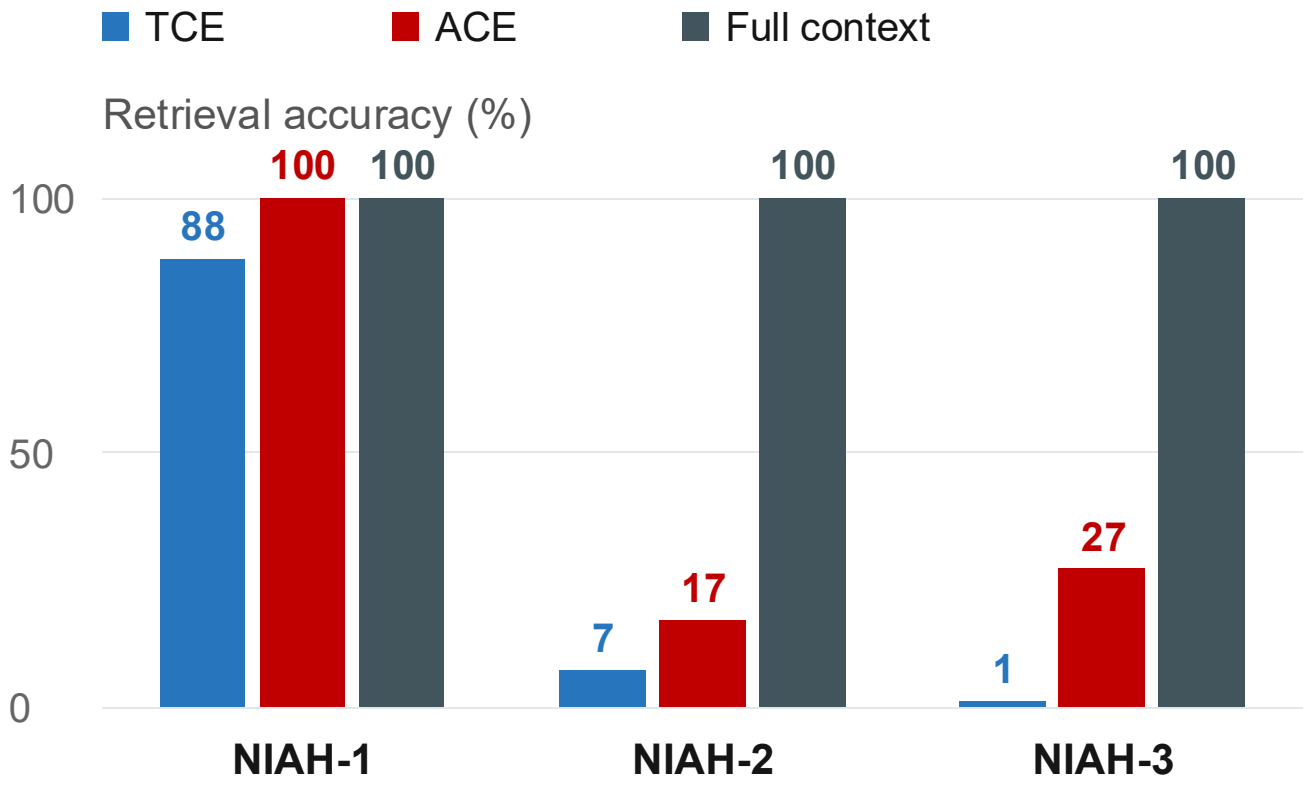
```
the grass is green. the sky is blue.
the access code is 4827193
the grass is green. the sky is blue.
```

NIAH-2 7-digit number in essay text

```
... an unrelated passage continues ...
the access code is 4827193
... more unrelated sentences follow ...
```

NIAH-3 UUID in essay text

```
... an unrelated passage continues ...
ID: 6c38a1e2-…-91ed21
... more unrelated sentences follow ...
```



ACE improve TCE, but difficult retrieval still trails full context.

~1K-token contexts · Qwen3.5-9B

LongBench

Document question answering after context encoding.

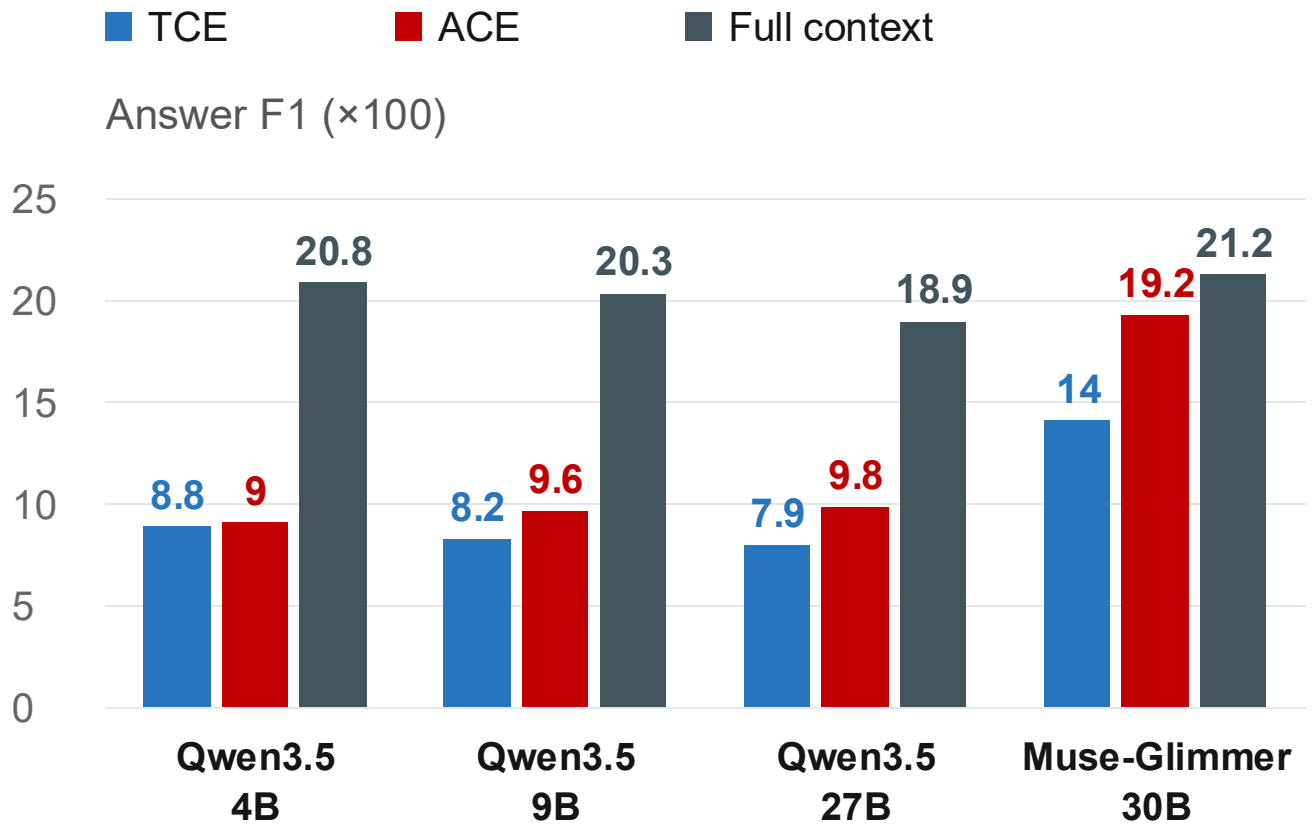
Data

The museum closed for renovation in 2016.
The project restored the main gallery and expanded the exhibition space.
The museum reopened on 12 May 2018.
Visitors could again explore the collection.

Question

When did the museum reopen?

Answer: 12 May 2018



ACE improves TCE, but still trails full context.

2K–7K-token documents · 100 QA instances

ENCODING CONTEXT INTO LLM WEIGHTS: WHAT MAKES IT CONTEXT?

Yue Wang[1] **Qizhou Wang**[2] **Feng Liu**[1] **Gang Niu**[2] **Masashi Sugiyama**[2,3] **Bo Han**[4]

[1] The University of Melbourne [2] RIKEN Center for Advanced Intelligence Project
[3] The University of Tokyo [4] TMLR Group, Hong Kong Baptist University

https://arxiv.org/abs/2610.17686

ENCODING CONTEXT INTO LLM WEIGHTS:
WHAT MAKES IT CONTEXT?

Thank you for listening!

Yue Wang[1] Qizhou Wang[2] Feng Liu[1] Gang Niu[2] Masashi Sugiyama[2,3] Bo Han[4]

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